

Graph knowledge tracing in cognitive situation: Validation of classic assertions in cognitive psychology

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ABSTRACT

Knowledge Tracing (KT) is a fundamental and challenging task in intelligent education, aiming to trace learners' knowledge states and learning processes, providing better support and guidance for teaching and addressing mental factors. Previous KT tasks have focused on considering learners' exposure to extrinsic environmental factors while ignoring the influence of intrinsic psychological factors. Moreover, previous methods have adopted a single perspective in modeling learners' knowledge states, ignoring the diversity of states in the learning process. To address these issues, we define the concept of *cognitive situation* through the guidance of cognitive psychology theory to help to explain the extrinsic influence and intrinsic cognition of learners within complex learning environments. Moreover, we design a Cognitive Situation-based Graph KT (CSGKT) model to quantify learners' influences in the cognitive process by modeling schemas capturing intrinsic characteristics and extrinsic factors through Hyper-Graph Neural Networks (HGNN). Second, we utilize a Directed Graph Convolutional Neural Network (DGCNN) to capture the correlation information between knowledge concepts and structure the learner's cognitive activities and knowledge states, adding a detailed representation of multiple states of the learning process. In addition, we use the Erase-add Gate to filter out the knowledge states that do not match the learner's current cognitive activities to stabilize the learner's due cognition. In our experiments, we selected nine baseline models from three mainstream approaches, including sequence-based approaches, *Transformer*-based approaches, and complex structure-based approaches. The experimental results show that our models outperform these baseline models. At the same time, we also verify two classic assertions in cognitive psychology, namely, the "short-term memory forgetting of knowledge concepts is mainly caused by interference rather than memory trace fading" and the "cognitive imagery and perceptual function play an equivalent role in the cognitive process", which further support the feasibility of the model.

1. Introduction

The rapid development of artificial intelligence and big data technologies has made intelligent education a key innovation in the educational field. Technologies [1] such as Knowledge Tracing (KT) provide personalized learning [2,3] support through real-time analysis of learners' behaviors and knowledge states, aiming to improve learning efficiency and educational quality by optimizing resource allocation. However, a major challenge in achieving intelligent education is accurately understanding learners' cognitive processes in complex learning environments. Learning is a social process embedded in specific contexts, influenced by external factors and individual characteristics. Ignoring this complexity hinders a comprehensive understanding of learners' cognitive states and impairs personalized support.

Education is inherently a complex system [4], encompassing numerous intricate factors such as interactions between individuals and groups, mutual shaping of environment and cognition, and the integration and analysis of multi-source heterogeneous data. A holistic understanding of learning environments necessitates interdisciplinary theoretical exploration and methodological innovation. Cognitive psychology provides a crucial theoretical foundation for comprehending cognitive processes in learning. According to cognitive psychologist Beck [5], human cognitive activities invariably unfold within specific situations, with situational factors such as learning environment, presentation of learning materials, and types of learning tasks influencing cognitive processing, including attention, memory, and thinking. In other words, learners are embodied, contextualized cognitive subjects.

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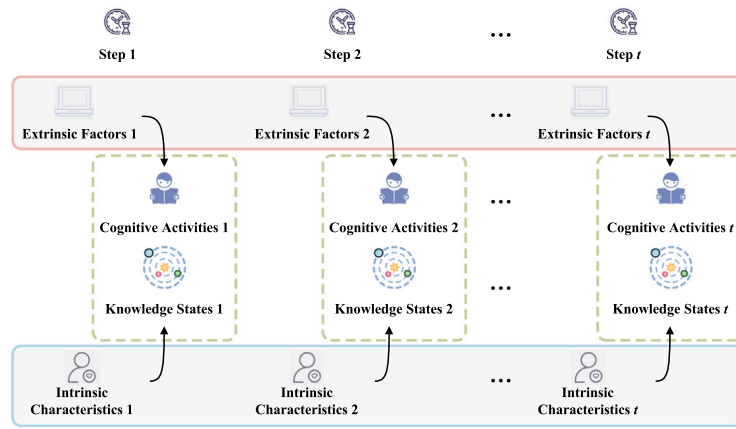


Fig. 1. Learners' cognitive activities and knowledge states at each time step are influenced by extrinsic factors and intrinsic characteristics, respectively.

Consequently, neglecting that situation precludes an accurate grasp of the essence of the learning process.

In recent years, the field of cognitive psychology has witnessed a proliferation of studies exploring cognitive engagement and cognitive states, offering new perspectives for understanding cognitive dynamics in learning processes. Liu et al. [6] employed deep learning methods to identify learners' cognitive states, revealing e-learning patterns under different cognitive engagement modes. Some studies focus on analyzing the associations between cognitive factors and learning outcomes, such as predicting academic performance based on emotions and cognitive engagement [7], and the impact of cognitive interference and emotional patterns on academic achievement [8]. Additionally, some scholars have proposed novel approaches like semi-supervised learning for identifying cognitive patterns in online learning discussions, achieving favorable results [9]. These studies expand the application of cognitive modeling in intelligent education; however, they still lack a systematic consideration of the multi-dimensional influencing factors in complex learning environments.

To overcome the limitations of existing KT methods, this paper introduces the concept of *cognitive situation*. A cognitive situation specifically refers to a learner's cognitive processing in a particular spatiotemporal situation, emphasizing the dynamic associations among extrinsic factors, intrinsic characteristics, cognitive activities, and knowledge states, thereby establishing a contextualized learner cognitive model. This concept provides a theoretical framework for analyzing key elements and their interaction patterns in learning environments. Within a cognitive situation, it is posited that a learner's knowledge state not only updates but also be influenced by intrinsic characteristics such as individual differences, perceptions of task difficulty, knowledge reception capacity, and adaptation to teaching methods, leading to variations in knowledge states. Fig. 1 illustrates that at each time step, both knowledge states and cognitive activities change to varying degrees due to the influence of extrinsic factors and intrinsic characteristics. By focusing on cognitive situations, a more comprehensive understanding of learners' cognitive change patterns in complex learning environments can be achieved. This approach supports the design of personalized and effective learning strategies, thereby enhancing learners' learning outcomes and mastery levels.

Our specific contributions are described as follows:

- By leveraging insights from cognitive psychology and typical application scenarios, we propose defining the cognitive situation and developing a correlation analysis model based on “extrinsic factors-cognitive process-intrinsic characteristics”. This model aims to enhance the alignment of KT tasks with real-world situations.
- CSGKT captures the schema of intrinsic features and extrinsic factors through HGNN modeling, and DGCNN is modeled to capture the correlation information between knowledge concepts and Erase-add Gate to rationalize learners' cognition.
- Extensive experiments were conducted on four benchmark datasets and compared with nine baseline models from three mainstream approaches. The experimental results show that CSGKT achieves the most advanced prediction performance.
- Two major experiments in cognitive psychology were simulated to validate the two classic assertions: “short-term memory forgetting of knowledge concepts is mainly caused by interference rather than memory trace fading” and “cognitive imagery and perceptual functions play an equivalent role in the cognitive process”.

Organizational Structure: The organizational structure for the rest of this article is as follows. Section 2 presents the background information, while Section 3 elaborates on the KT problem and introduces the proposed CSGKT model. Section 4 explains the experimental setup, and Section 5 presents the experimental results and analysis. Based on the KT results, Section 6 discusses the results of two simulation verification experiments for our approach. Section 7 is the conclusion. The paper ends with the discussion in Section 8.

2. Related work

In this section, we begin with an extensive review of the three dominant approaches. We then outline the specific definitions that give cognitive situations, as they are closely related to the work in this paper.

2.1. Knowledge tracing

KT [10,11] as a fundamental task in intelligent education, possesses a wide range of potential and has come to prominence in recent decades. It involves continuous monitoring, assessment of learners' knowledge and skills, and tracing of the knowledge state. This tracing not only helps to understand the individual needs and development of learners but can also be used to adjust educational strategies to better meet the learning needs of learners. When it comes to the high performance and timeliness of KT research, it can usually be categorized into three main types of approaches: sequence-based approaches, *Transformer*-based approaches, and complex structure-based approaches. Each of these approaches has different advantages and applicability, and the appropriate model can be selected to improve the performance and efficiency of KT based on specific research and educational scenarios.

2.1.1. Sequence-based approaches

Neural networks are commonly used in the field of KT due to their superior ability to learn from sequential data, allowing them to effectively process large amounts of information. In comparison to traditional models, neural networks have significantly improved performance in this area. The Deep KT (DKT) model [12] utilizes a Recurrent Neural Network (RNN) to extract learners' potential knowledge proficiency from their motion sequences. However, DKT has limitations such as relying on a single input, which can be addressed by incorporating various rich features [13,14] of learners to individualize and account for differences between them [15]. The Learning and Forgetting behavior modeling for KT (LFKT) [16] model not only considers the forgetting factor but also combines learners' mastery of each knowledge concept and their responses in real time to trace their learning progress. Another major limitation of DKT is that it cannot judge the mastery of a particular knowledge concept. The Dynamic Key-Value Memory Networks for KT (DKVMN) [17] model uses dynamic and static matrices to store and update learners' mastery level of each knowledge concept, addressing this limitation. DKVMN is a memory network [18] that refers to the use of a dynamic matrix to address changing conditions in real time, while the static matrix stores more permanent information. The recent Question Difficulty Consistent KT (QDCKT) model [19] innovatively combines Long Short-Term Memory (LSTM) [20,21] networks and feed-forward neural networks, replacing question IDs with question difficulty levels. This approach facilitates the transformation of model predictions into estimations of learners' skill knowledge states. Sequence-based models rely on a single input, failing to capture the multidimensional nature of learning, which leads to an oversimplified representation of learners' knowledge states, thereby affecting the effectiveness of learning and prediction accuracy. A study by Yeung et al. [22] demonstrated that such simplifications can result in up to a 15% reduction in prediction accuracy when dealing with complex, interdisciplinary subjects. Moreover, these models do not account for individual learning styles and cognitive processes, which can undermine the effectiveness of personalized learning strategies. Research by Zhang et al. [23] showed that adaptive learning systems based solely on sequence data failed to improve learning outcomes for up to 30% of students, particularly those with diverse learning needs. Therefore, it is crucial to develop more comprehensive and adaptable models to better capture the complex dynamics of the learning process.

2.1.2. Transformer-based approaches

The *Transformer* [24] has been utilized in learners' studying tasks. It incorporates a self-attention mechanism to enhance its ability to capture intrinsic relationships and interactions [25–27]. One of the most representative models is the Attentive KT (AKT) model [25], which has achieved advanced performance in KT tasks. Additionally, Pu et al. proposed a KT model that combines DKT and *Transformer* models, called DKT+*Transformer* [28]. This model utilizes the correlation between time intervals and knowledge concepts as input. The SAINT and SAINT+ models [29,30] suggest that current KT models based on attention mechanisms do not fully utilize deep self-attention mechanisms to capture the complex relationship between questions and answers. To address this, a new input method is adopted, and the use of attention layers is increased to improve the experimental results. HiTSKT [31] model employs a hierarchical *Transformer* architecture to capture both intra-session and inter-session knowledge dynamics. Additionally, it incorporates a power-law decay attention mechanism to simulate learners' forgetting behavior.

2.1.3. Complex structure-based approaches

Based on the Graph-based KT (GKT) [32], the GKT-CD [33] introduces cognitive diagnostic techniques to overcome the limitations of the traditional KT approach in analyzing learners' abilities. It initially considered that learners would be affected by intrinsic characteristics, but did not explore them further. In Session Graph-based KT

(SGKT) [34], on the other hand, the concept of a session graph is used to model the learner interaction process. This interaction mechanism enables the knowledge states and embedded representations to interact with each other, further enhancing the comprehensive performance of the model. However, SGKT focuses more on the aspect of knowledge states being influenced by embedded representations and lacks multiple perspectives. Various methods have been proposed in the KT task to utilize graphic structures and relational information [35–37]. Yang et al. [38] introduced a Graph-based Interaction model for KT (GIKT), which employs a Graph Convolutional Network (GCN) to merge question-skill correlation through embedding. Abdelrahman et al. [39] extracted the relationship between potential concepts from the dynamic potential concept map according to knowledge state changes. Ni et al. [40] input the question concept dichotomous graph directly into the graph embedding module to obtain the representation of each question and concept. They used the representation of each question and concept and the learners' learning history to encode their engagement and knowledge state. Sun et al. [41] utilized question-question collaborative embedding technology to embed similarity-based learning problems for integrated operation as the input of the neural network. Song et al. [42] established links between questions under cross-concept to help capture high-level semantic information. Tong et al. [43] structured hierarchical question diagrams to model question-learning dependency. Gan et al. [44] integrated the difficulty of cognitive problems to obtain a comprehensive representation of each question and enhanced the acquisition of question and skill embedding in the graph. Cui et al. [45] established the dual graph structure of learners' learning interaction and, on this basis, jointly captured the heterogeneous movement concept association and interactive transformation. The Contrastive Learning for KT (CL4KT) [46] model introduces contrastive learning to enhance model performance by learning more discriminative feature representations. The HawkesKT [47] model incorporates the Hawkes process to model temporal cross-effects in KT, capturing the cumulative impact of historical interactions and their temporal evolution. This approach provides a novel perspective for modeling the dynamics of knowledge acquisition. Although these methods effectively capture comprehensive information, they only take into account the evolution of the knowledge state and produce unexpected deviations from the cognitive situation.

2.2. Cognitive situation

In the KT task, we define the cognitive situation as “a complex interplay between learners' extrinsic factors, intrinsic cognitive characteristics, and the learning process, which profoundly affects an individual's state knowledge and cognitive activity [5]”. This conceptualization draws inspiration from extensive research in cognitive psychology and educational science, endeavoring to synthesize perspectives from Vygotsky's sociocultural theory [48] and Piaget's cognitive development theory [49]. It reflects the multidimensional and dynamic nature of the learning process. This integrative approach not only encapsulates the influence of learning environments on cognitive development but also encompasses the evolution of internal cognitive structures, thus providing a more comprehensive theoretical foundation for understanding the complexities of learning. To illustrate the practical advantages of the cognitive situation concept, consider a real-world scenario in an online learning platform. Traditional KT models might only track a student's performance on math problems, adjusting difficulty based on correct or incorrect answers. In contrast, a system employing the cognitive situation concept would also consider factors such as the time of day the student is studying (an extrinsic factor that might affect concentration), the student's recent performance in related subjects (intrinsic characteristic reflecting overall academic state), and the specific cognitive processes involved in problem-solving (e.g., spatial reasoning vs. algebraic manipulation). For instance, if the system detects that a typically high-performing student is struggling with geometry problems

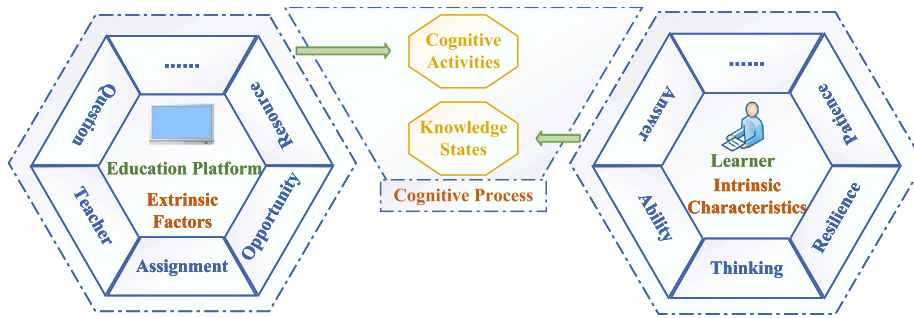


Fig. 2. An “extrinsic factors-cognitive process-intrinsic characteristics” correlation analysis model.

late in the evening, it might infer fatigue rather than lack of knowledge, and suggest a break or switch to a different topic. This nuanced approach could lead to more effective learning interventions and more accurate predictions of long-term learning outcomes.

The concept of cognitive situation emphasizes that learning is a complex process involving the interplay of multiple factors, rather than a mere accumulation of knowledge. This aligns with contemporary learning science perspectives, such as the “How People Learn” framework proposed by Bransford et al. [50] and the situated learning theory of Lave and Wenger [51]. These theories collectively construct a multi-layered model of learning, underscoring the critical roles of environment, prior knowledge, motivation, and sociocultural situation in the learning process. They provide a robust theoretical underpinning for the conceptualization of cognitive situation.

The complex composition of the cognitive situation covers several important dimensions. In Fig. 2, we visualize the correlation analysis model “extrinsic factors-cognitive process-intrinsic characteristics” for cognitive situations. This model specifies the interrelationships between the elements of the cognitive situation, and the green arrows indicate the unidirectional influence of the elements. Extrinsic factors, i.e., the organization of the physical environment and social interactions, can influence information acquisition and cognitive processes, including questions, teachers, assignments, opportunities, resources, etc., and have a clear impact on learners’ cognitive activities. The significance of these extrinsic influences is thoroughly explained in Bandura’s social cognitive theory [52], which emphasizes the dynamic interplay among individual cognition, behavior, and environmental factors. Intrinsic characteristics, which cover the learner’s characteristics, include answers, abilities, thinking, resilience, and patience. These intrinsic characteristics not only reflect learners’ personalities but also influence their performance in cognitive tasks. Dweck’s Mindset Theory [53] and Duckworth’s research on grit [54] further explain the mechanisms through which these intrinsic characteristics influence learning outcomes.

Cognitive processes are fundamental mechanisms through which humans perceive the extrinsic world and process information [4]. These processes reflect the dynamic changes in cognitive states, which comprise cognitive activities (such as attention, perception, memory, and reasoning) and knowledge states (including mastered concepts, skills, and experiences). These cognitive activities and knowledge states continuously interact and update during the learning process, forming complex cognitive processes.

The model design in the cognitive situation is informed by two key cognitive psychology assertions. Firstly, the assertion that “short-term memory forgetting of knowledge concepts is mainly caused by interference rather than memory trace fading” emphasizes the importance of inter-concept interactions during the learning process. In the CSGKT model, the DGCNN component is designed to capture complex relationships between different knowledge concepts, simulating potential interference and facilitation effects among concepts in actual learning scenarios. This design reflects our understanding of short-term memory

forgetting mechanisms and underscores the necessity of considering dynamic interactions between concepts in KT.

Secondly, the assertion that “cognitive imagery and perceptual functions play an equivalent role in the cognitive process” posits that intrinsic cognitive representations and extrinsic perceptual inputs play equally important roles in the learning process. Our model addresses this through the Heterogeneous Graph Neural Network (HGNN) component, which simultaneously processes learners’ intrinsic characteristics (analogous to cognitive representations) and extrinsic factors (analogous to perceptual functions), integrating them into a unified framework. This design reflects our consideration of the balanced role of intrinsic representations and extrinsic stimuli in cognitive processes. By incorporating these cognitive psychology theories into the model design, we attempt to create a KT system that is not only technically feasible but also more theoretically sound.

3. Proposed method

In Section 2.2, we outline the conceptual framework of the cognitive situation. This theoretical construct serves to simulate real learning environments, including the complex interactions among extrinsic factors, intrinsic characteristics, and cognitive processes that collectively shape a learner’s knowledge acquisition trajectory. This framework, rooted in cognitive psychology theories such as Vygotsky’s sociocultural theory and Piaget’s cognitive development theory, addresses a key limitation of existing KT methods: their inability to account for the multifaceted nature of learning environments. The CSGKT model functions as an instrumental tool for capturing, quantifying, and simulating the dynamic evolution of this cognitive situation. By employing various effective components, CSGKT transforms the abstract concept of the cognitive situation into measurable parameters and predictive algorithms. For instance, the use of HGNN to model extrinsic factors and intrinsic characteristics allows the model to consider the complex interplay between environmental stimuli and learners’ cognitive processes — an aspect often overlooked in traditional KT approaches. This approach facilitates a more nuanced and comprehensive analysis of the learning process, transcending the limitations of traditional KT methods that often focus on isolated learning instances. Furthermore, the incorporation of DGCNN to model knowledge concept relationships reflects the cognitive psychology principle of knowledge organization and transfer, addressing the shortcomings of existing methods that treat knowledge items as independent entities. Through the integration of theoretical foundations and computational modeling, we bridge the gap between idealized representations of learning environments and the practical challenges of KT in real-world educational settings.

This section describes the KT task and delineates the operationalization of the CSGKT model. Fig. 3 illustrates the comprehensive architectural framework of the CSGKT model. This framework employs four distinct modules to execute the KT task. The first module is to form a schema. The inputs are the set of extrinsic factor nodes V^z and the set of intrinsic characteristic nodes V^f . This module uses the HGNN to form the graphs for the extrinsic factors (*Factor A*, ..., *Factor N*)

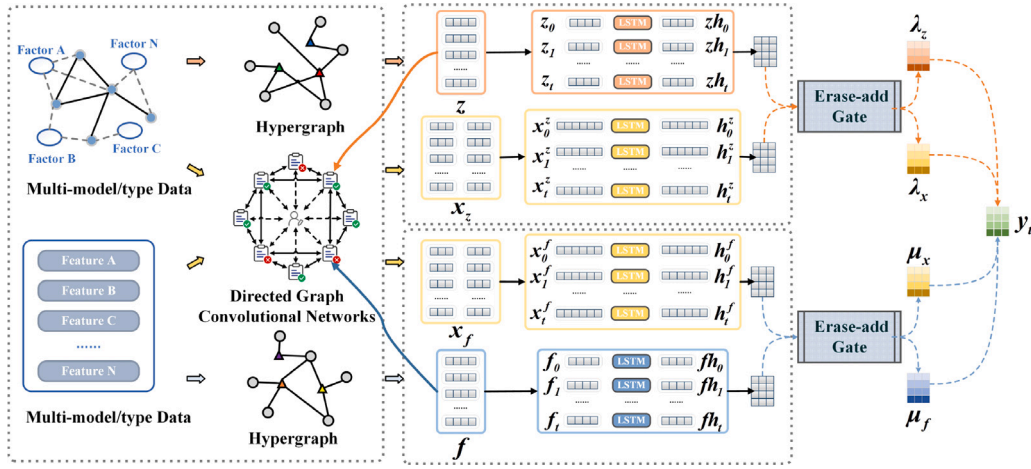


Fig. 3. An overview of the overall architecture of the CSGKT model.

and intrinsic characteristics (*Feature A*, ..., *Feature N*) are quantized to obtain the extrinsic factor pattern z and intrinsic feature pattern f , respectively. z and f denote the combined pattern of extrinsic factor and intrinsic characteristic influences on the learner, respectively. The second module is to form the associated information. The input is the set of knowledge concepts C , which consists of a blend of question attributes and skill attributes from extrinsic factors, and answer attributes from intrinsic characteristics. This module uses DGCNN to learn the association between knowledge concepts and obtains the cognitive activities x_z , which reflects the learner's state in the learning process, and the knowledge states x_f , which reflects the learner's mastery of knowledge. The third module is state evolution. The inputs are the four parameters obtained from the first and second modules. This module uses LSTM to train the temporal hidden states of the four parameters and obtains zh_t , fh_t , h_t^z , and h_t^f . The fourth module is stable cognition. The input is the hidden state obtained from the third module. This module uses the Erase-add Gate mechanism to filter out cognitive bias and match the cognitive activity with the knowledge state to get the final prediction result y_t . Next, we describe these four modules in detail. The pseudo-code of CSGKT is described as Algorithm 1.

3.1. Problem formulation

In an online education platform, each question q_i corresponds to one or more knowledge concepts. Learners acquire the relevant knowledge concepts by answering the questions. Let $Q = \{q_1, q_2, \dots, q_n\}$ denote the question set containing n questions, where q_i denotes the i th question, which is a k -dimensional one-hot vector, i.e., $q_i \in \mathbb{R}^k$. The learner's (question, answer) sequence is denoted as $S = \{(q_1, a_1), (q_2, a_2), \dots, (q_t, a_t)\}$, where $q_i \in Q$ denotes the question answered by the learner at timestamp i , and t is the most recent timestamp. Each learner can answer the question of interest at a specific time and give a correct or incorrect response, denoted as $a_i \in \{0, 1\}$. If $a_i = 0$, the answer is wrong; if $a_i = 1$, the answer is correct. Given a learner's past learning sequence S and a new question q_{t+1} , we need to model the learner's historical learning sequences to trace the changes in their knowledge state over time. Our goal is to predict the probability $P(a_{t+1} = 1 \mid S, q_{t+1})$ that the learner will answer the new question correctly based on the current knowledge state.

3.2. Forming schemas

HGNN [55] extends the concept of traditional GNN to handle hypergraph data. In HGNN, both nodes and hyperedges of the hypergraph can carry multiple elements or attributes, and the connectivity relationships of the hyperedges can be more complex. The key idea of HGNN

Algorithm 1 Cognitive Situation-based Graph Knowledge Tracing (CSGKT)

Input: Extrinsic factor nodes set V^z , Intrinsic characteristic nodes set V^f , Knowledge concept set C , Interaction sequence S

Output: Predicted result y_t

```

1: initialize node embeddings for  $V^z$  and  $V^f$ 
2: initialize hyperedge embeddings for hyperedges connecting nodes
   in  $V^z$  and  $V^f$ 
3: for node in  $V^z$  do
4:    $V^z\_embedding \leftarrow Embedding(node, hyperedges\_connected\_to\_node)$ 
5: end for
6: for node in  $V^f$  do
7:    $V^f\_embedding \leftarrow Embedding(node, hyperedges\_connected\_to\_node)$ 
8: end for
9:  $z \leftarrow hypergraph(V^z\_embedding)$  ▷ Equation (3)
10:  $f \leftarrow hypergraph(V^f\_embedding)$  ▷ Equation (4)
11: initialize directed edges based on the knowledge concept set  $C$ 
12: for node in  $C$  do
13:    $C\_embedding \leftarrow Embedding(node, directed\_edges)$ 
14: end for
15:  $x_z \leftarrow directed\_graph\_convolution(C\_embedding, z)$  ▷ Equation (8)
16:  $x_f \leftarrow directed\_graph\_convolution(C\_embedding, f)$  ▷ Equation (8)
17: for  $t$  in range(len( $S$ )) do
18:    $zh[t], fh[t], h^z[t], h^f[t] \leftarrow LSTM(z), LSTM(f), LSTM(x_z),$ 
      $LSTM(x_f)$  ▷ Equation (9)
19: end for
20:  $\lambda_z, \lambda_x, \mu_x, \mu_f \leftarrow EraseAdd\_Gate(zh_t, h_t^z, h_t^f, fh_t)$  ▷ Equation (10)
21:  $y_t \leftarrow fusion(\lambda_z, \lambda_x, \mu_x, \mu_f)$  ▷ Equation (10)
22: return  $y_t$ 

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is to capture the relationships between the nodes in the hypergraph by designing appropriate hypergraph convolution operations. These convolution operations consider all the elements in the hyperedges to compute the representation of the nodes. Traditional graph structures struggle to express the multifaceted relationships in cognitive situations. For instance, factors such as “question difficulty”, “learner ability”, and “learning environment” collectively influence a learner's knowledge state and cognitive level, a multi-dimensional association that binary edges struggle to capture. The hyperedges in HGNNs, capable of connecting multiple nodes, are well-suited for modeling such higher-order relationships.

Schema refers to the cognitive process of abstracting and summarizing basic information about a specific kind of object and forming a mental framework diagram. In HGNN, we can create several groups

Table 1

Parameter names and parameter explanations for the dataset corresponding to the ten extrinsic factors and seven intrinsic characteristics.

Type	Annotation	Parameter	Explanation
Extrinsic Factors	Question	problem_id	The ID of the problem.
	Assignment	assignment_id	All content associated with a problem is called an assignment and has the same assignment ID.
	Teacher	status_id	Used to tell if a teacher has marked an assignment as excused.
	Class	learner_class_id	The ID of the class.
	Network	network_state	The state of the network when the learner did the assignment.
	School	school_id	ID of the school where the problem was assigned.
	Opportunity	opportunity	The number of opportunities the learner has to practice on this skill.
	Homework	homework_percent	The percent of learners who did the assignment as homework (number of assignments/number of assignments finished).
	Resource	skill_id	ID of the skill associated with the problem
	Tutor	tutor_mode	This column indicates the mode the learner is in, including tutor, test mode, pre-test, or post-test.
Intrinsic Characteristics	Answer	correct	This column is the target for prediction.
	Ability	user_id	The ID of the learner doing the problem.
	Thinking	first_response_time	Time between start time and first learner action (asking for hint or entering an answer).
	Self-discipline	bottom_hint	This column is the prompt for the question and contains the answer to the question.
	Resilience	overlap_time	The time in milliseconds for the learner to complete the problem.
	Adaptability	first_action	This column indicates the learner's first reaction to seeing the problem, including attempt, hint, scaffolding or learner clicked on the problem but did nothing else.
	Patience	attempt_count	Number of attempts(number of times a learner entered an answer).

of hyperedge structures by utilizing the intricate correlations among multi-modes/multi-data in extrinsic factors and intrinsic characteristics. Through the “node-edge-node” (NEN) transformation, we can efficiently extract high-order schema on the hypergraph. Specifically, suppose there are m extrinsic factor nodes $V^z = \{v_1^z, v_2^z, \dots, v_m^z\}$ and n intrinsic characteristic nodes $V^f = \{v_1^f, v_2^f, \dots, v_n^f\}$, where v_i^z denotes the i th m -dimensional extrinsic factor node and v_j^f denotes the j th n -dimensional intrinsic characteristic node. Table 1 lists the specific information of extrinsic factor nodes and intrinsic characteristic nodes. We integrate learning factors into the hypergraph structure through extrinsic factor nodes V^z and intrinsic characteristic nodes V^f . To construct the hypergraph structure, a similarity function $S(v_i, v_j)$ is first defined to measure the association strength between nodes:

$$S(v_i, v_j) = \cos(v_i, v_j) = \frac{v_i \cdot v_j}{\|v_i\| \|v_j\|} \quad (1)$$

where $\cos$ represents cosine similarity. Subsequently, a threshold τ is set, and a hyperedge is created between nodes v_i and v_j when $S(v_i, v_j) > \tau$. This approach allows for the capture of complex multidimensional relationships, extending beyond simple binary relations.

Extrinsic factor nodes V^z correspond to extrinsic factors influencing the learning environment, encompassing ten aspects including questions, assignments, teachers, classes, and networks. These factors influence cognitive processes by stimulating learners' cognitive activities. For instance, different tutoring modes require varying levels of cognitive resources, while question difficulty and relevant skills affect learners' information processing demands. Factors such as networks and homework percentage also impact learners' learning outcomes.

Intrinsic characteristic nodes V^f represent learners' traits and cognition, comprising seven aspects including answer correctness, ability, thinking time, and self-discipline. These intrinsic characteristics influence cognitive processes by shaping learners' knowledge states

and, more importantly, reflect individual differences among learners. For example, ability (user_id) can be viewed as a representation of learners' prior knowledge and overall learning capacity; thinking time (first_response_time) and initial action (first_action) reflect learners' cognitive processing speed and question-solving strategies, revealing different learning styles; self-discipline (bottom_hint) and patience (attempt_count) be related to learning motivation, demonstrating learners' persistence and help-seeking behaviors when facing challenges. Through these multidimensional intrinsic characteristics, our model can capture individual differences in learning styles, motivation levels, and knowledge foundations, thereby providing a more personalized learning experience for each learner.

To better capture the relationships between these nodes, we use HGNN to describe the associations beyond the individual nodes. Therefore, we define an Extrinsic Factor Hyper-Graph (EFHG), i.e. $G^z = (V^z, E^z, W^z)$ and an Intrinsic Characteristic Hyper-Graph (ICHG), i.e. $G^f = (V^f, E^f, W^f)$, where E^z and E^f are the hyperedge set of EFHG and ICHG respectively. We initialize W^z and W^f as diagonal matrices of hyperedge weights, with dimensions of $|E^z|$ and $|E^f|$, i.e., $W^z = I_{|E^z|}$ and $W^f = I_{|E^f|}$, respectively. Based on the similarity function in Eq. (1), the incidence matrices $H^z \in \mathbb{R}^{|V^z| \times |E^z|}$ and $H^f \in \mathbb{R}^{|V^f| \times |E^f|}$ for the EFHG and ICHG can be defined as follows:

$$H^z(i, j) = \begin{cases} 1, & \text{if } S(v_i^z, v_j^z) > \tau, \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

$$H^f(i, j) = \begin{cases} 1, & \text{if } S(v_i^f, v_j^f) > \tau, \\ 0, & \text{otherwise} \end{cases}$$

here, a hyperedge e^z is formed when the similarity between v_i^z and v_j^z exceeds τ , and the set of these hyperedges constitutes E^z . Similarly, a hyperedge e^f is formed when the similarity between v_i^f and v_j^f exceeds τ , and the set of these hyperedges constitutes E^f . This construction

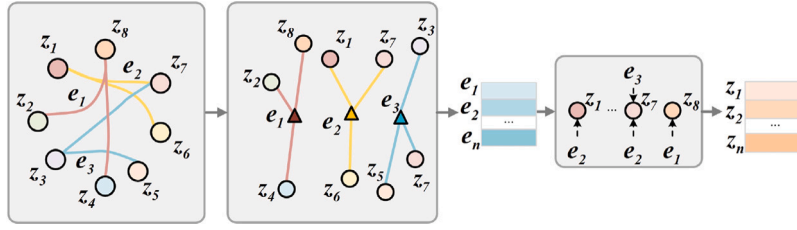


Fig. 4. The extrinsic factor node's essential information transforms.

method ensures that the hypergraph captures higher-order similarity relationships between nodes.

To acquire the essential information contained within node v_i^z , we utilize embedding techniques to obtain the extrinsic factor schema z_i . We then leverage the capabilities of HGNN to establish the rules governing the propagation of information through the convolution operator at each layer. z_i can be dynamically updated through the process of aggregating the relevant information emanating from both node v_i^z and its neighboring nodes, as per Eq. (3).

$$z_i^{(l+1)} = \delta((D_v^z)^{-1/2} H^z W^z (D_e^z)^{-1/2} (H^z)^T (D_v^z)^{-1/2} z_i^l \Theta^l) \quad (3)$$

where $z_i^{(l+1)}, z_i^l \in \mathbb{R}^m$, denote the input embedding and output embedding of the extrinsic factor node v_i^z between the two layers, respectively. $\Theta^l \in \mathbb{R}^{m \times m}$ is a learnable parameter matrix of layer l and the propagation matrix within the node layer. $H^z \in \mathbb{R}^{m \times |E^z|}$, the node-hyperedge association matrix, represents the connectivity of the nodes and hyperedges. $W^z \in \mathbb{R}^{|E^z| \times |E^z|}$ is the hyperedge weight matrix. $\delta(\cdot)$ represents a nonlinear activation function such as LeakyReLU [56]. $D_v^z \in \mathbb{R}^{m \times m}$, is the degree matrix of the extrinsic factor node v_i^z , and the diagonal elements denote the degree of v_i^z . $D_e^z \in \mathbb{R}^{|E^z| \times |E^z|}$, is the degree matrix of the extrinsic factor hyperedges, and the diagonal elements denote the degree of the hyperedges. They denote the number of connections of nodes and hyperedges, and the characteristic aggregation of nodes and hyperedges is smoothed by taking the inverse of the number of nodes and hyperedges. Where $D_v^z = \sum_{e_j} W^z(e) H_{ij}^z (v_i \in V^z, e_j \in E^z)$ and $D_e^z = \sum_{v_i} H_{ij}^z (v_i \in V^z, e_j \in E^z)$.

In Fig. 4, the extrinsic factor node's essential information transforms. This is achieved through the use of the HGNN layer, which facilitates NEN transformations, thereby allowing for a more refined treatment of extrinsic factors and intrinsic characteristics. Specifically, the initial node's essential information z_i^l is filtered using a learnable processing matrix Θ^l , and the node's essential information is gathered based on the hyperedge, resulting in the creation of the hyperedge feature H^z . By multiplying the filtered information and the hyperedge feature, the essential information of the output node is obtained by aggregating the essential information of the relevant hyperedge. Consequently, the HGNN layer can be effectively utilized to extract the higher-order schema from both the EFHG and ICHG through NEN transformations. This enhances the overall efficacy of the approach.

Similarly, we can update the intrinsic characteristic schema f_i by aggregating information from the neighboring nodes of v_i^f and its associated hyperedge using Eq. (4).

$$f_i^{(l+1)} = \delta((D_v^f)^{-1/2} H^f W^f (D_e^f)^{-1/2} (H^f)^T (D_v^f)^{-1/2} f_i^l \Omega^l) \quad (4)$$

where $f_i^{(l+1)}, f_i^l \in \mathbb{R}^n$, denote the input embedding and output embedding of intrinsic characteristic node v_i^f between the two layers, respectively. $\Omega^l \in \mathbb{R}^{n \times n}$ is a learnable parameter matrix of layer l and the propagation matrix within the node layer. $H^f \in \mathbb{R}^{n \times |E^f|}$, the node-hyperedge association matrix, represents the connectivity of the nodes and hyperedges. $W^f \in \mathbb{R}^{|E^f| \times |E^f|}$ is the hyperedge weight matrix. $D_v^f \in \mathbb{R}^{n \times n}$, is the degree matrix of the intrinsic characteristic node v_i^f , and the

diagonal elements denote the degree of v_i^f . $D_e^f \in \mathbb{R}^{|E^f| \times |E^f|}$, is the degree matrix of the intrinsic characteristic hyperedges, and the diagonal elements denote the degree of the hyperedges. They denote the number of connections of nodes and hyperedges, and the feature aggregation of nodes and hyperedges is smoothed by taking the inverse of the number of nodes and hyperedges. Where $D_v^f = \sum_{e_j} W^f(e) H_{ij}^f (v_i \in V^f, e_j \in E^f)$ and $D_e^f = \sum_{v_i} H_{ij}^f (v_i \in V^f, e_j \in E^f)$.

3.3. Forming associated information

DGCNN [57] exploits directional information in directed graphs for node representation learning and graph classification tasks. The core idea of DGCNN is to aggregate the neighbor information of the nodes through a convolutional operation, which considers the direction of the edges. This allows DGCNN to distinguish between incoming and outgoing edges and thus better capture the topology of the directed graph. It uses a convolutional layer to process the neighbor information of the nodes to generate a representation of the nodes. Moreover, relationships between knowledge concepts are typically directional. For instance, "addition" is a prerequisite for "multiplication", but not vice versa. GCNs disregard edge directions, potentially leading to confusion in information propagation. DGCNN preserves edge direction information, allowing knowledge concept representations to aggregate along correct paths, thus more accurately depicting knowledge structures.

In Section 3.1, we established that each question must involve at least one knowledge concept, denoted as $c_i \in C$, where $C = \{c_1, c_2, \dots, c_k\}$ represents a set of knowledge concepts of size k . Specifically, the set of knowledge concepts C is a combination of question attributes and skill attributes that are extrinsic factors, and answer attributes that are intrinsic features. Question attributes and skill attributes correspond to each other, and they reflect the connection between the question and the corresponding skill point, i.e., the expression of the knowledge concept. The answer attribute, on the other hand, reflects the accuracy of the answer. These attributes are interwoven to form a directed graph to establish relationships between knowledge concepts and make them more visible. To construct a directed graph of knowledge concepts, a relationship strength function $R(c_i, c_j)$ is defined as follows:

$$R(c_i, c_j) = P(c_j | c_i) = \frac{\text{count}(c_i \rightarrow c_j)}{\text{count}(c_i)} \quad (5)$$

where $\text{count}(c_i \rightarrow c_j)$ represents the frequency of c_j immediately following c_i in the learning sequence, and $\text{count}(c_i)$ denotes the total occurrences of c_i . This function measures the transition probability from concept c_i to c_j .

In knowledge systems, concepts inherently possess precedence or succession relationships to varying degrees. Consequently, each knowledge concept c_i is decoupled into an in-degree node c_i^{in} and an out-degree node c_i^{out} , transforming the knowledge concept set C into a directed structure $C' = \{c_1^{\text{in}}, c_1^{\text{out}}, c_2^{\text{in}}, c_2^{\text{out}}, \dots, c_k^{\text{in}}, c_k^{\text{out}}\}$ with in-degrees and out-degrees. As a result, the size of the new knowledge concept set C' becomes $2k$. This directed structure is defined as $G_c = (C', E_c)$,

where $E_c \subseteq C' \times C'$ represents the edge set. For any two original concepts c_i and c_j , if a relationship exists from c_i to c_j , a directed edge is added from c_i^{out} to c_j^{in} in the new graph structure. Learners' mastery of knowledge concepts is influenced not only by historical learning information S but also by extrinsic factors and intrinsic characteristics. Based on the relationship strength function in Eq. (5), the decoupled adjacency matrix $A \in \mathbb{R}^{2k \times 2k}$ is defined as follows:

$$A_{ij} = \begin{cases} R(c_{[i/2]}, c_{[j/2]}) & \text{if } i \% 2 = 0 \text{ and } j \% 2 = 1 \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

where $[i/2]$ represents the floor division and $i \% 2$ represents the modulo operation. This definition ensures that only edges from out-degree nodes to in-degree nodes have non-zero weights, maintaining the directionality of relationships between knowledge concepts. As defined in Eq. (7), the directed edges are normalized to obtain the transition adjacency matrix $A' \in \mathbb{R}^{2k \times 2k}$, which assigns weights to each directed edge:

$$A'_{ij} = \frac{A_{ij}^p}{\sum_k A_{ik}^p} \quad (7)$$

where p represents the norm. This normalization method ensures that the sum of outgoing edge weights for each node equals 1 while preserving directional information of the edges. Note that this normalization applies to nodes with non-zero out-degrees.

Based on Eq. (8), the association information between nodes is extracted using DGCNN to derive the initial cognitive activity x_z and knowledge state x_f of learners.

$$\begin{aligned} x_z^{(l+1)} &= [\delta(D^{-1/2} A' D^{-1/2} z_i^l \Phi^l), \delta(D^{-1/2} A'^T D^{-1/2} z_i^l \Phi^l)] \\ x_f^{(l+1)} &= [\delta(D^{-1/2} A' D^{-1/2} f_i^l \Psi^l), \delta(D^{-1/2} A'^T D^{-1/2} f_i^l \Psi^l)] \end{aligned} \quad (8)$$

Since the dimensions m and n of the node features are much larger than the dimension k of the knowledge concepts, the dimension $d = \min\{m, n, 2k\}$ is defined here. $D \in \mathbb{R}^{d \times d}$ is the degree matrix, $D_{ii} = \sum_j (A'_{ij} + A'_{ji})$. $z_i^l, f_i^l \in \mathbb{R}^{d \times d}$ are the node features of the previous layer. $\Phi, \Psi \in \mathbb{R}^{d \times d}$ are the model parameters to be learned in the network.

3.4. State evolution

KT is fundamentally a sequence modeling problem, where learners' cognitive states change dynamically over time. In educational scenarios, this change typically exhibits both long-term trends (such as gradual improvement in abilities) and short-term fluctuations (such as variations in attention). Numerous studies [12,58,59] have demonstrated that LSTM networks are well-suited for sequence prediction problems with long-term dependencies. The gating mechanisms and memory cell design of LSTM make it particularly adept at capturing both long-term and short-term dependencies in sequential data, making it highly suitable for modeling the evolution of cognitive states.

Thus, we adopt LSTM to model the dynamic changes in extrinsic factor schema z , intrinsic characteristic schema f , cognitive activity x_z , and knowledge state x_f over time. During each training, it is necessary to trace the hidden state of each parameter at the t th timestamp, which is denoted as zh_t, fh_t, h_t^z , and h_t^f . Eq. (9) illustrates the updating process, taking the cognitive activity as an example.

$$\begin{aligned} i_t &= \sigma(x_t^z W_{x_z,i} + h_{t-1}^z W_{h_z,i} + b_i) \\ f_t &= \sigma(x_t^z W_{x_z,f} + h_{t-1}^z W_{h_z,f} + b_f) \\ o_t &= \sigma(x_t^z W_{x_z,o} + h_{t-1}^z W_{h_z,o} + b_o) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tanh(x_t^z W_{x_z,c} + h_{t-1}^z W_{h_z,c} + b_c) \\ h_t^z &= o_t \odot \tanh(c_t) \end{aligned} \quad (9)$$

where, i_t, f_t , and o_t represent the input gate, forget gate, and output gate, respectively. The variable c_t is the cell state vector, while $W_* \in \mathbb{R}^{d \times d}$ and $b_* \in \mathbb{R}^d$ are trainable network parameters. The function $\sigma(\cdot)$ represents a non-linear activation function, such as *Sigmoid*.

3.5. Stable cognition

In cognitive processes, learners' cognitive activities and knowledge states are not always in perfect alignment. For instance, learners sometimes overestimate their abilities (high cognitive activity with a low knowledge state) or underestimate themselves due to frustrations (low cognitive activity with a high knowledge state). Simply merging these two states lead to prediction biases. As observed in Fig. 3, various parameters operate independently, advancing according to specific rules within their respective domains, potentially resulting in mismatches between learners' knowledge states and cognitive activities.

Inspired by the literature [17], an Erase-add Gate is designed to address this mismatch issue. The core concept of the Erase-add Gate is to align learners' current knowledge states and cognitive activities by dynamically adjusting the contributions of different situational factors. This mechanism effectively resolves the mismatch problem by allowing the model to dynamically "erase" irrelevant information and "add" important new information at each time step. Specifically, when a learner's cognitive activity level exceeds their actual knowledge state, the Erase-add Gate reduces the weight of cognitive activity while increasing the influence of the knowledge state, thereby avoiding overly optimistic predictions. Conversely, when a learner's knowledge state surpasses their cognitive activity performance, the mechanism appropriately enhances the contribution of cognitive activity to prevent the model from underestimating the learner's potential.

This dynamic adjustment process simulates selective attention and information update mechanisms in human cognition, enabling the model to more accurately reflect changes in learners' cognitive states across different learning stages. Specifically, the operation of the Erase-add Gate is shown in Eq. (10).

$$\begin{aligned} [\lambda_z, \lambda_x, \mu_x, \mu_f] &= [\xi_{ea}(zh_t), \xi_{ea}(h_t^z), \xi_{ea}(fh_t^f), \xi_{ea}(fh_t)] \\ y_t &= \sigma(\lambda_x - \sigma(\lambda_z + \lambda_x)\lambda_z + \tanh(\mu_x + \mu_f)\mu_x \mu_f) \end{aligned} \quad (10)$$

here, $\xi_{ea}(\cdot)$ is the Erase-add Gate function, defined as:

$$\xi_{ea}(x) = \sigma(W_e x + b_e) \odot \tanh(W_a x + b_a) \quad (11)$$

where $W_e, W_a \in \mathbb{R}^{d \times d}$ are weight matrices, $b_e, b_a \in \mathbb{R}^d$ are bias vectors, and d is the dimension of the hidden state. $\odot$ denotes element-wise multiplication, σ is the sigmoid function, and $\tanh$ is the hyperbolic tangent function. $\sigma(W_e x + b_e)$ controls the degree of erasing, while $\tanh(W_a x + b_a)$ generates newly added information. $\lambda_z, \lambda_x, \mu_x$, and μ_f are new states obtained after filtering extrinsic factor patterns, intrinsic characteristic patterns, cognitive activities, and knowledge states through the Erase-add Gate.

In Eq. (10), the first term λ_x represents the current cognitive activity state, the second term $\sigma(\lambda_z + \lambda_x)\lambda_z$ simulates the "erasing" effect of extrinsic factors on cognitive activities, while the third term $\tanh(\mu_x + \mu_f)\mu_x \mu_f$ represents the fusion of knowledge states and cognitive activities, i.e., the process of "adding" new information. This design enables the model to dynamically adjust the balance between knowledge states and cognitive activities based on the current cognitive situation, thereby more accurately capturing the learner's actual cognitive processes.

Finally, the model is optimized using the cross-entropy loss function. More specifically, we calculate the difference between the ground truth answer a_t and the predicted result y_t , while accounting for the impact of extrinsic factors, intrinsic characteristics, cognitive activity, and knowledge state on the model. The cross-loss term between the true answer and the predicted result is shown in Eq. (12).

$$l(a_t, y_t) = -\frac{1}{n} \sum_{i=1}^n (\zeta(a_t) \log \zeta(y_t) + (1 - \zeta(a_t)) \log (1 - \zeta(y_t))) \quad (12)$$

where, n represents the total number of questions, and $\zeta(\cdot)$ denotes that the nonlinear activation function is performed twice. Similarly, we can obtain the cross-entropy loss term for other influencing factors $l(a_t, \lambda_z)$,

Table 2
Statistics of four datasets used in this study.

Datasets	Assist2009	Assist2017	Statics2011	KDDCup2010
# of learners	4 151	686	333	574
# of questions	16 891	4 117	1 223	661
# of concepts	110	102	80	436
# of interaction records	325 673	942 816	189 927	607 026
Avg. records per learner	78.46	1374.37	570.35	1057.54
Avg. questions per concept	153.55	40.36	15.29	1.52

$l(a_t, \lambda_x)$, $l(a_t, \mu_x)$, $l(a_t, \mu_f)$, and add L2-regularization to the objective function, as shown in Eq. (13).

$$\lambda = \|\zeta(\lambda_x) - \zeta(y_t)\|_2 + \|\zeta(\lambda_x) - \zeta(y_t)\|_2 + \|\zeta(\mu_x) - \zeta(y_t)\|_2 + \|\zeta(\mu_f) - \zeta(y_t)\|_2 \quad (13)$$

The parameter λ is a balancing hyperparameter that controls the relative contribution of the cross-entropy loss and the impact factor loss. To achieve higher accuracy of the prediction results during training, the objective function is shown in Eq. (14).

$$Loss = l(a_t, y_t) + l(a_t, \lambda_x) + l(a_t, \lambda_x) + l(a_t, \mu_x) + l(a_t, \mu_f) + \lambda \quad (14)$$

4. Experimental setting

In this section, we introduced four datasets and nine models that need to be compared and described the details of the experimental settings.

4.1. Datasets

To ensure a comprehensive evaluation of our model, we selected four publicly available datasets spanning multiple years: ASSISTments 2009, KDDCup2010, Statics2011, and ASSISTments2017. These datasets represent distinct temporal nodes in the evolution of educational technology and learning paradigms. They encompass extensive interaction records between learners and authentic computer-based tutoring systems across different periods. This temporal diversity renders them ideal for research in large-scale, long-term dynamic KT environments. Moreover, these datasets cover various educational domains with varying difficulty levels, providing a robust foundation for assessing the generalizability of our research findings. Table 2 presents detailed statistics of these datasets, highlighting their unique characteristics and temporal contexts.

ASSISTments2009(Assist2009)¹ was collected during the 2009–2010 school year and contained a total of 525,535 interactions in its first release, which included duplicates that were prone to unreliable results. In the experiments of this paper, an updated version [17] is used, which contains datasets with two different interaction modes: the “skill-builder” and the “non-skill-builder”. The “skill-builder” dataset is used in the experiments because it fixes the data modeling problem and removes duplicate records. The updated dataset contains 4151 learners, 110 question labels, and 325,637 interactions. This dataset primarily focuses on secondary school mathematics, encompassing topics such as algebra, geometry, and number sense. The difficulty levels range from foundational to intermediate. The distribution of correct and incorrect answers exhibits a right-skewed distribution, indicating that the majority of questions have a relatively high rate of correct answers.

ASSISTments2017(Assist2017)² comes from the 2017 data mining competition on the ASSISTment platform. It contains 4117 different questions answered by 1709 learners over 102 knowledge concepts,

generating 942,816 interactions. As part of the data mining competition, this dataset contains the most descriptive information from all the task datasets. Similar to Assist2009, this dataset also covers secondary school mathematics but encompasses a broader range of difficulty levels, spanning from foundational to advanced topics. The distribution of correct and incorrect answers more closely approximates a normal distribution, indicating a more uniform distribution of question difficulties.

Statics2011³ comes from an engineering statics course at Carnegie Mellon University in Fall 2011. The original dataset consisted of 361,092 interactions, 335 learners, and 1224 questions. In the KT literature, this dataset is usually preprocessed [17] to obtain 1223 different questions. A total of 333 learners answered questions on 80 knowledge concepts. After preprocessing, the number of learner interactions was reduced by almost half to 189,297. This preprocessing was because most of the interactions lacked information about whether the question was answered correctly or not. The preprocessing only considered the first valid attempt. This dataset represents a more advanced domain, focusing on university-level engineering statics. The problems are typically more challenging, with the distribution of correct and incorrect answers exhibiting a left-skewed distribution, indicating that many questions are relatively difficult for learners. The dataset encompasses concepts such as force systems, equilibrium, and structural analysis.

KDDCup2010⁴ is a data provided by Carnegie Corporation on learners generated in the math course, 607,026 interactions on 436 questions for 574 learners, containing detailed steps for each question and corresponding mastery of knowledge concepts. This dataset encompasses a broad spectrum of high school mathematics topics, including algebra, geometry, and trigonometry. The difficulty levels exhibit significant variation, with the distribution of correct and incorrect answers displaying a bimodal pattern. This bimodality reflects a polarization in question difficulty, encompassing both relatively simple questions and considerably challenging ones. The dataset is particularly rich in step-by-step question-solving information, enabling a more nuanced analysis of learners’ learning processes.

4.2. Compared models

The selection of baseline models adhered to a comprehensive set of criteria, ensuring a rigorous and equitable evaluation of CSGKT. We prioritized models that demonstrate diverse methodological approaches in the KT domain, encompassing sequence-based models (DKT, DKVMN, QDCKT), Transformer-based models (SAKT, AKT, HiTSKT), and complex structure-based models (GKT, HawkesKT, CL4KT). The chosen models reflect a spectrum from seminal works that have shaped the trajectory of KT research to state-of-the-art implementations, enabling a robust assessment of CSGKT’s performance within both historical and contemporary frameworks.

Furthermore, we focused on models that share conceptual or structural affinities with CSGKT, such as graph-based architectures or those leveraging attention mechanisms, facilitating nuanced comparisons of

¹ <https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data/skill-builder-data-2009-2010>

² <https://sites.google.com/view/assistmentsdataminingdataset?authuser=0>

³ <https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=507>

⁴ <https://pslcdatashop.web.cmu.edu/KDDCup/downloads.jsp>

specific model components. The selected models have demonstrated superior performance in previous empirical studies, ensuring that CSGKT is benchmarked against high-performance standards in the field.

DKT [12] represents a significant milestone in KT as the first model to apply Recurrent Neural Networks (RNN) to this field. Its selection is crucial for establishing a baseline for neural network approaches in KT. DKT's use of one-hot skill encoding and hidden vector representations provides a fundamental comparison point for more complex models, allowing us to assess the advancements made since its introduction.

DKVMN [17] was chosen for its innovative use of dynamic and static matrices to model learner knowledge. This model addresses a key limitation in earlier approaches by explicitly tracking mastery levels of individual knowledge concepts. Its inclusion enables us to evaluate the effectiveness of memory-augmented neural networks in KT tasks.

SAKT [60] marks the introduction of attention mechanisms in KT. Its selection is vital for understanding how attention-based models improve upon sequence models like DKT. SAKT's ability to weight the importance of items in long sequences offers insights into handling long-term dependencies in learning trajectories.

AKT [25] represents a sophisticated integration of attention mechanisms with psychometric models. We included AKT to assess the benefits of incorporating domain-specific knowledge (through the Rasch model) into neural architectures. Its novel monotonic attention mechanism provides a unique perspective on modeling forgetting in learning processes.

GKT [32] was selected as a representative of graph-based approaches in KT. Its use of Graph Neural Networks (GNN) to model skill dependencies aligns closely with our CSGKT model's graph-based components. Comparing CSGKT with GKT allows us to evaluate the advancements made in graph-based knowledge representation.

HawkesKT [47] introduces point processes to model temporal dynamics in KT. Its inclusion enables us to compare our approach with models that explicitly focus on capturing complex temporal patterns in learning. This comparison is crucial for assessing CSGKT's ability to model time-dependent aspects of learning.

CL4KT [46] represents the application of contrastive learning in KT. We selected this model to evaluate how CSGKT compares with approaches that leverage advanced feature learning techniques. This comparison helps in assessing the robustness of our model's feature representations.

QDCKT [19] was chosen for its explicit consideration of question difficulty in learner performance modeling. This model allows us to compare how CSGKT handles varying task complexities without explicitly modeling question difficulty, providing insights into our model's adaptability to diverse learning materials.

HiTSKT [31] employs a hierarchical temporal self-attention mechanism, representing an advanced approach to modeling long-term dependencies. Its inclusion allows us to assess CSGKT's performance in capturing knowledge state changes over extended periods, a critical aspect of comprehensive KT.

Through comparison with these models, we can comprehensively evaluate our proposed model's capabilities in capturing the complexity of learning processes, handling temporal information, and utilizing knowledge structures. Simultaneously, this multi-dimensional comparison aids in accurately identifying our innovations and contributions, while validating the model's effectiveness and novelty in KT tasks.

4.3. Setup and implementation

Before the experiment, the training datasets and test datasets are split by extracting 80% of the sequences from the datasets as the former and reserving the remaining 20% as the latter.

We embedded the problem concept representation into a 256-dimensional space to obtain the optimal hyperparameters. For HGNN and DGCNN, a two-layer GNN was employed for node embedding, while the LSTM model utilized a hidden layer and 128 hierarchical

cells. The remaining models adopted the settings described in the original paper or used the AUC values obtained on the validation datasets.

In the HGNN, the optimal similarity threshold τ was determined to be 0.6 through grid search. For the DGCNN, a 3-layer structure was adopted, effectively capturing multi-hop relationships without overfitting. The normalization parameter p for the relationship strength function was experimentally set to 2, preserving important relationships while smoothing extreme values. These parameters achieved the best balance on the validation set, ensuring model robustness without excessively altering the original graph structure.

Across the four datasets, the model was optimized using Adam optimizer with an initial learning rate of 0.001, employing a learning rate decay strategy of 10% every 10 epochs. Unlike existing models, the input sequence length was fixed at 50. Long sequences were divided into multiple subsequences, while short sequences were padded with empty symbols to reach the fixed length.

An early stopping strategy was implemented during training, halting the process when the AUC on the validation set showed no improvement for 5 consecutive epochs. The model was tested for 50 epochs, as it converged around that period. To ensure result stability, each experiment was repeated 5 times. Optimal parameter settings for other baselines were as specified in their original works.

Predicting learners' responses to questions was defined as a binary classification task. As shown in Table 2, interaction records exhibited imbalance due to correct and incorrect answers. Consequently, the Area Under the Receiver Operating Characteristic (ROC) Curve (AUC) was chosen as the primary evaluation metric, widely used in existing research. Higher AUC values indicate better performance in predicting learner responses, with 0.5 representing random guess performance. Additionally, Accuracy (ACC), Root Mean Square Error (RMSE), and F1-score were employed as supplementary evaluation metrics. ACC reflects the model's overall accuracy, RMSE quantifies the average deviation between predicted and actual values, sensitive to larger errors, and F1-score provides a balanced view of model performance in potentially imbalanced class scenarios. These metrics are highly relevant to the KT task. In predicting learners' mastery of specific knowledge points, AUC assesses the model's ability to distinguish between mastered and unmastered knowledge points, crucial for adaptive learning system decisions. ACC directly reflects the accuracy of predicting learner performance, aligning with educational assessment standards. RMSE helps identify significant deviations in model predictions, particularly useful for understanding model performance on questions of varying difficulty. F1-score is especially important in addressing potential imbalances in correct answer rates, ensuring good model performance in predicting both correct and incorrect responses.

5. Results and analysis

In this section, we report a series of experimental results to validate the effectiveness of the proposed CSGKT.

5.1. Performance prediction

The capability of different models in predicting learner performance was evaluated through experimental results on four datasets: Assist2009, Assist2017, Statics2011, and KDDCup2010. Table 3 presents detailed performance metrics for all models, including AUC, ACC, RMSE, and F1-score. The exceptional performance of the CSGKT model on these metrics not only reflects its predictive accuracy but also aligns with multiple factors in cognitive situations.

On the Assist2009 dataset, the CSGKT model achieved an AUC value of 0.8515 and an accuracy of 0.8068. These high metrics reflect the model's excellent ability to distinguish between learners who have mastered and not mastered knowledge points. The F1-score of 0.7879 indicates that the model has achieved a good balance between precision

Table 3

Comparisons of the results of different models on the four datasets.

Dataset	Metric	DKT	DKVMN	SAKT	AKT	GKT	HawkesKT	CL4KT	QDCKT	HiTSKT	CSGKT
Assist2009	AUC	0.7337	0.7398	0.7498	0.8346	0.7962	0.7629	0.8337	0.7889	0.8142	0.8515
	ACC	0.7534	0.7646	0.7582	0.7749	0.7665	0.8125	0.8168	0.8027	0.7543	0.8068
	RMSE	0.4965	0.4562	0.4440	0.4235	0.4249	0.4038	0.4013	0.4164	0.3985	0.4164
	F1	0.7514	–	–	–	0.7543	0.7212	0.7885	0.7432	0.7261	0.7879
Assist2017	AUC	0.7070	0.7073	0.6648	0.7702	0.7189	0.7682	0.769	0.7711	0.7553	0.7734
	ACC	0.6687	0.6572	0.6800	0.6886	0.6701	0.7404	0.7425	0.7143	0.6845	0.6943
	RMSE	0.4469	0.4392	0.4475	0.4263	0.4718	0.4557	0.4544	0.4289	0.4387	0.4188
	F1	0.4776	0.4200	0.5200	0.5477	0.5231	0.6131	0.7538	0.6124	0.6131	0.7532
Statics2011	AUC	0.8233	0.8195	0.8029	0.8265	0.7343	0.7301	0.7943	0.8225	0.8012	0.8542
	ACC	0.7650	0.7647	0.7540	0.7650	0.7214	0.7463	0.7651	0.7541	0.7421	0.785
	RMSE	0.4056	0.3975	0.4003	0.3945	0.4136	0.4215	0.3945	0.4107	0.4003	0.3594
	F1	0.7200	–	–	–	0.7102	0.6874	0.7962	0.7254	0.7134	0.8558
KDDCup2010	AUC	0.7510	0.7530	0.8042	0.8152	0.7691	0.7465	0.8097	0.7624	0.7653	0.8158
	ACC	0.6840	0.6868	0.7280	0.6956	0.7024	0.6942	0.7201	0.6892	0.7126	0.7268
	RMSE	0.5528	0.5596	0.5215	0.5517	0.5321	0.4563	0.3526	0.4395	0.4124	0.3438
	F1	0.8050	–	–	–	0.7865	0.6231	0.8589	0.4395	0.7341	0.8714

and recall, which is equally important for identifying learners who need additional help and those who have already grasped concepts.

The model obtained the highest AUC value of 0.8542 and ACC of 0.785 on the Statics2011 dataset. Statics2011 represents a statics course in higher education, and CSGKT's higher predictive performance reflects more stable and predictable learning patterns exhibited by college learners in professional course learning. This aligns with the typically stronger self-directed learning abilities and more focused learning attitudes of higher education learners. While DKVMN uses discrete key-value memory to store static and dynamic information, CSGKT offers a more comprehensive approach by simultaneously modeling the interplay between evolving curricular content (extrinsic factors) and learners' developing problem-solving competencies (intrinsic characteristics) in relation to progressively complex curricular content. This dynamic interaction modeling enables educators to not only identify the onset of learning difficulties but also to understand their underlying causes—whether they stem from knowledge gaps, necessitate alternative instructional approaches, or result from temporal changes in learners' cognitive strategies.

As mentioned earlier, our analysis also reveals the significant impact of intrinsic characteristics on learners' knowledge states. For instance, on the Statics2011 dataset, learners' ability (user_id) and thinking time (first_response_time) showed strong correlations with their performance, with correlation coefficients of 0.72 and -0.56 respectively. This indicates that learners with higher ability tend to have more stable and advanced knowledge states, while longer thinking times often correlate with less confident knowledge states.

In contrast, on the Assist2017 dataset, the model's AUC value was 0.7734 with an ACC of 0.6943, relatively lower. This reflects the complexity of primary and secondary education environments. At this stage, learners' studies are influenced by more situational factors, increasing the difficulty of prediction. Simultaneously, the lower RMSE value indicates that despite challenges, the model's prediction errors remain relatively small. This performance corresponds to the actual situation in primary and secondary education, where learners' learning trajectories tend to be more variable.

The model's performance was equally outstanding on the KDDCup2010 dataset, with an AUC of 0.8158 and ACC of 0.7268. This dataset had the lowest RMSE value (0.3438), indicating more precise predictions by the model. This reflects a more structured educational environment represented by this dataset, with more efficient learning behaviors among the learner population.

These results not only demonstrate the model's technical performance but, more importantly, reflect its adaptability to different educational stages and learning environments. For instance, the model performs best on college courses (Statics2011) while showing relatively

weaker performance on datasets covering a broader age range (Assist2017). This discrepancy aligns with reality: as learners age and accumulate learning experiences, their learning processes tend to become more stable and predictable.

Furthermore, the model's performance across different metrics also reflects the challenges and needs in actual education. Higher AUC values indicate excellent model performance in distinguishing learners of different levels, which is particularly valuable for implementing stratified teaching strategies. Lower RMSE values suggest that the model's predictions are generally accurate, which is especially important for precise allocation of educational resources and development of personalized learning plans.

5.2. Model comparisons

This section presents a comprehensive comparison between the CSGKT model and selected baseline models. Our analysis encompasses performance metrics across four datasets, statistical significance tests, and the impact of input variations. We evaluate the models using AUC, ACC, RMSE, and F1 scores, and conduct paired t-tests to assess the significance of CSGKT's improvements. Furthermore, we investigate how different input configurations affect model performance and examine the influence of CSGKT's structural design on its efficacy.

5.2.1. Baseline comparison

The CSGKT model generally outperforms baseline models across the four datasets, albeit with subtle differences in certain metrics. A comprehensive analysis of AUC, ACC, RMSE, and F1-score is conducted to explore CSGKT's advantages and potential areas for improvement.

In terms of AUC, CSGKT demonstrates superior performance across all datasets, showing improvements of 2.02%, 0.42%, 3.35%, and 0.07% respectively compared to the best baseline models. This consistency reflects CSGKT's robust ability to distinguish between learners' mastered and unmastered knowledge points, attributed to its comprehensive consideration of learning environment dynamics, knowledge structures, and learning processes.

However, CSGKT's performance in ACC is more complex. It achieves the highest accuracy on Statics2011 and KDDCup2010 datasets. Yet, on Assist2009 and Assist2017 datasets, CSGKT's ACC is slightly lower than some baseline models like CL4KT. This is because CSGKT, in pursuing more comprehensive modeling, introduces excessive complexity in certain simple prediction tasks. For instance, more concise models have an advantage when dealing with relatively simple problems.

Regarding RMSE, CSGKT achieves the lowest values on Assist2017, Statics2011, and KDDCup2010 datasets, indicating minimal prediction errors. However, on the Assist2009 dataset, CSGKT's RMSE is slightly higher compared to QKCKT, reflecting a minor overfitting phenomenon

Table 4
Paired t-test Results of CSGKT against Baseline Models (AUC Metric)

		Assist2009	Assist2017	Statics2011	KDDCup2010
DKT	<i>t</i>	15.32,	12.18	8.76	10.54
	<i>p</i>	<0.001	<0.001	<0.001	<0.001
	<i>d</i>	3.42	2.72	1.96	2.36
	CI	[0.102, 0.134]	[0.056, 0.077]	[0.024, 0.041]	[0.054, 0.076]
DKVMN	<i>t</i>	14.87	11.93	9.12	9.87
	<i>p</i>	<0.001	<0.001	<0.001	<0.001
	<i>d</i>	3.32	2.67	2.04	2.21
	CI	[0.097, 0.127]	[0.054, 0.074]	[0.028, 0.046]	[0.050, 0.071]
SAKT	<i>t</i>	13.21	14.56	11.34	3.76
	<i>p</i>	<0.001	<0.001	<0.001	<0.01
	<i>d</i>	2.95	3.25	2.53	0.84
	CI	[0.087, 0.116]	[0.093, 0.120]	[0.043, 0.064]	[0.005, 0.017]
AKT	<i>t</i>	5.43	1.98	7.89	0.32
	<i>p</i>	<0.001	0.062	<0.001	0.752
	<i>d</i>	1.21	0.44	1.76	0.07
	CI	[0.012, 0.023]	[0.000, 0.006]	[0.021, 0.036]	[-0.003, 0.004]
GKT	<i>t</i>	11.76	10.21	18.43	9.65
	<i>p</i>	<0.001	<0.001	<0.001	<0.001
	<i>d</i>	2.63	2.28	4.12	2.16
	CI	[0.046, 0.065]	[0.043, 0.062]	[0.107, 0.134]	[0.037, 0.055]
HawkesKT	<i>t</i>	16.54	2.87	20.12	12.32
	<i>p</i>	<0.001	<0.01	<0.001	<0.001
	<i>d</i>	3.70	0.64	4.50	2.75
	CI	[0.078, 0.098]	[0.002, 0.009]	[0.111, 0.137]	[0.058, 0.081]
CL4KT	<i>t</i>	6.21	2.13	13.76	2.54
	<i>p</i>	<0.001	<0.05	<0.001	<0.05
	<i>d</i>	1.39	0.48	3.07	0.57
	CI	[0.012, 0.024]	[0.000, 0.007]	[0.051, 0.070]	[0.001, 0.010]
QDCKT	<i>t</i>	14.32	1.89	8.43	11.21
	<i>p</i>	<0.001	0.074	<0.001	<0.001
	<i>d</i>	2.20	0.42	1.88	2.51
	CI	[0.054, 0.075]	[0.000, 0.005]	[0.024, 0.040]	[0.044, 0.064]
HiTSKT	<i>t</i>	9.87	7.32	12.54	10.76
	<i>p</i>	<0.001	<0.001	<0.001	<0.001
	<i>d</i>	2.21	1.64	2.80	2.4
	CI	[0.030, 0.045]	[0.013, 0.023]	[0.045, 0.063]	[0.041, 0.060]

when handling certain types of data. Cognitive situation modeling combines extrinsic factors, such as curricular content and resource availability, with intrinsic characteristics, including prior knowledge and learning styles. This integration provides a more comprehensive framework for addressing heterogeneous and dynamic learning environments. For instance, in the Assist2017 dataset, which represents primary and secondary education, CSGKT demonstrates adaptability to the rapidly evolving learning environments of younger learners. It takes into account factors such as fluctuations in assignment difficulty and the development of learners' self-management skills. Consequently, the model more accurately predicts performance variations across diverse subjects and learning stages. This capability distinguishes CSGKT from models like CL4KT, which primarily focus on characteristic representation without considering the full spectrum of situational factors.

In F1 score, CSGKT performs best on Statics2011 and KDDCup2010 datasets, demonstrating a good balance between precision and recall. However, on Assist2009 and Assist2017 datasets, CSGKT's F1 scores are slightly lower than the CL4KT model. This is due to these datasets containing more diverse problem types and learner groups, increasing the difficulty of accurate predictions. CSGKT consistently maintains superior F1 scores across diverse datasets, as shown by its performance on Assist2009 (0.7879 compared to GKT's 0.7543). This demonstrates CSGKT's balanced approach in identifying both mastered and non-mastered concepts. Such balance surpasses graph-based models like GKT, which occasionally overemphasize structural relationships at the expense of crucial characteristics.

CSGKT's superior performance in capturing the influence of intrinsic characteristics on knowledge states is particularly evident. For example, unlike models such as DKT or DKVMN which primarily focus on

sequential patterns, CSGKT successfully modeled the impact of learners' self-discipline (bottom_hint) on their knowledge state evolution, revealing a moderate positive correlation (0.43) with performance improvement over time across all datasets.

CSGKT achieves multi-dimensional, dynamic modeling of the learning process by integrating HGNN, DGCNN, LSTM, and Erase-Add Gating units. This complex structure gives CSGKT an advantage in capturing cognitive states and learning process dynamics, especially when dealing with complex learning environments and knowledge structures. For example, HGNN allows the model to simultaneously consider learners' intrinsic characteristics and extrinsic factors, which is particularly important in professional course datasets like Statics2011, where learners' prior knowledge and learning environment significantly impact learning outcomes. The application of LSTM enhances the model's adaptability to dynamic learning processes, which is particularly advantageous in long-term learning datasets like KDDCup2010.

However, this structure also introduces unnecessary complexity in some simple scenarios, leading to slightly inferior performance in ACC and F1 metrics compared to simpler models. For instance, in middle school mathematics datasets like Assist2009 and Assist2017, where problem types are relatively simple and learner groups more homogeneous, relatively simpler models might perform better on certain metrics.

5.2.2. Statistical comparison

To rigorously evaluate the performance advantages of the CSGKT model relative to baseline models, comprehensive statistical significance tests were conducted. Paired *t*-tests were employed to compare the AUC performance of CSGKT against each baseline model across the

Table 5
The differences of the input to the modeling modular.

Model	Data Source			
	Questions	Knowledge Concepts	Context	Time
DKT_OQ	✓	✗	✗	✗
DKT_OKC	✗	✓	✗	✗
AKT	✓	✓	✓(question-answer)	✓(<i>Rasch</i> model)
AKT_RC	✓	✓	✗	✓(<i>Rasch</i> model)
AKT_RT	✓	✓	✓(question-answer)	✗
CSGKT	✓	✓	✓(attribution)	✓(attribution)
CSGKT_RKC	✓	✗	✓(attribution)	✓(attribution)
CSGKT_RC_RT	✓	✓	✗	✗

four datasets. AUC was selected as the primary evaluation metric due to its comprehensive reflection of model performance across different thresholds, unaffected by class imbalance. To ensure result reliability, 20 independent experiments were conducted, each initializing model parameters with different random seeds.

Table 4 presents the paired t -test results of CSGKT against various baseline models on the AUC metric, including t -statistics, p -values, Cohen's d effect sizes, and 95% confidence intervals for AUC differences. The interpretation standards for effect size d are: 0.2 indicates a small effect, 0.5 a medium effect, and 0.8 a large effect. Results demonstrate that CSGKT significantly outperforms baseline models in the majority of cases ($p < 0.05$), with 95% confidence intervals not including 0, further confirming the reliability of performance differences.

The results demonstrate that CSGKT significantly outperforms baseline models in the vast majority of cases ($p < 0.05$), with 95% confidence intervals excluding 0, further confirming the model's reliability.

On the Assist2009 dataset, CSGKT exhibits extremely high statistical significance and large effect sizes compared to all baseline models. For instance, against DKT, CSGKT's AUC improvement has a 95% confidence interval of [0.102, 0.134], indicating that in practical applications, CSGKT correctly identifies 10.2% to 13.4% more learners' knowledge states than DKT. This improvement in large-scale educational systems translates to thousands of learners receiving more precise learning guidance.

The Statics2011 dataset, representing engineering courses in higher education, showcases CSGKT's most significant advantages. Compared to GKT, the confidence interval for AUC improvement is [0.107, 0.134]. This substantial improvement indicates that CSGKT more accurately traces learners' knowledge mastery in complex engineering concept learning, crucial for adjusting the teaching pace, identifying learners needing additional support, and personalizing assignment allocation.

On the Assist2017 dataset, CSGKT's performance differences from some recent models (such as AKT and QDCKT) are less pronounced, with confidence intervals near 0. This reflects the complexity and diversity of primary and secondary education environments. For example, compared to AKT, the confidence interval is [0.000, 0.006], indicating comparable performance between the two models in certain scenarios. This underscores the need to select or combine different models based on specific educational stages and subject characteristics when designing adaptive learning systems.

Results on the KDDCup2010 dataset are particularly noteworthy. CSGKT shows significant advantages over most models, but the difference with AKT is not significant (CI: [-0.003, 0.004]). Given this dataset's characteristics (containing numerous algebra questions), this indicates that in highly structured subjects with clear logical relationships, different advanced models reach similar performance ceilings. Notably, CSGKT's ability to model the impact of intrinsic characteristics on knowledge states contributes to its statistical superiority. For instance, on the Statics2011 dataset, CSGKT's representation of the relationship between thinking time and knowledge state accuracy was significantly more precise than that of CL4KT ($p < 0.001$, $d = 3.07$), highlighting CSGKT's enhanced capability in modeling cognitive processes.

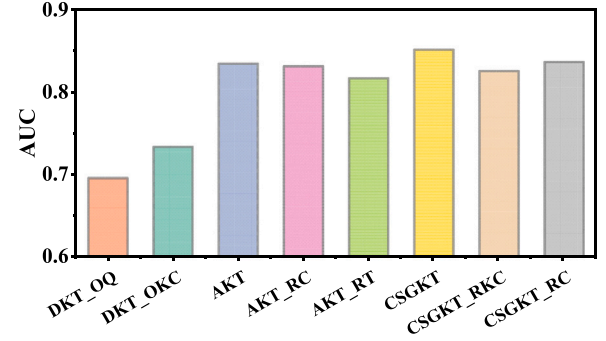


Fig. 5. AUC resulting from models with different inputs in the Assist2009 dataset.

Importantly, while CSGKT's advantages over some recent models (such as CL4KT and HiTSKT) are relatively small, they consistently achieve statistical significance. For instance, on the Statics2011 dataset, compared to CL4KT, the confidence interval for AUC improvement is [0.051, 0.070]. These continuous improvements are crucial for advancing educational technology, as even seemingly minor enhancements produce significant impacts in large-scale applications.

5.2.3. Input variation comparison

The input parameters of the CSGKT model include the set of extrinsic factor nodes V^z (containing question attributes and skill attributes), the set of intrinsic characteristic nodes V^f (containing answer attributes and time attributes), and the set of knowledge concepts C . These inputs are processed through different modules to form the extrinsic factor pattern z , intrinsic feature pattern f , cognitive activities x_z , and knowledge states x_f .

Considering that the inputs to the model also have an impact on the model performance, we compare eight models with different input parameters in Table 5. DKT_OQ denotes the DKT model with only the Question attribute, DKT_OKC denotes the DKT model with only the Knowledge Concepts attribute, and AKT_RC denotes the AKT model with the Context attribute removed, where the Context attribute refers to the context information extracted from the question-answer pairs. AKT_RT denotes the AKT model with the Time attribute removed, whereas the Time attribute refers to the *Rasch* model. CSGKT_RKC denotes the CSGKT model with the Knowledge Concepts attribute removed. CSGKT_RC denotes the CSGKT model with the Context attribute and Time attribute removed. CSGKT model with environmental factors removed, where Context attribute refers to extrinsic factors and intrinsic characteristics, which can be mentioned in Table 1, where the time attribute used by the CSGKT model is included in the intrinsic characteristics. The AUC values in the Assist2009 dataset are presented in Fig. 5. The DKT models (including DKT_OQ and DKT_OKC) represent a typical sequence modeling approach that focuses on the sequence of learners' answering history, and thus the dependence on the Question attribute and Knowledge Concepts attribute is quite

Table 6
Comparison of effectiveness of different components.

Model	Assist2009	Assist2017	Statics2011	KDDCup2010
CSGKT-RH	0.8366	0.7474	0.8378	0.8005
CSGKT-RD	0.8256	0.6789	0.7990	0.7940
CSGKT-RL	0.8275	0.7389	0.8285	0.7934
CSGKT-RE	0.8342	0.7311	0.7963	0.7589
CSGKT	0.8515	0.7734	0.8542	0.8158

significant. It mainly relies on learners' answer history to provide key clues to their knowledge mastery. Comparatively, the AKT models (including AKT_RC and AKT_RT) adopt the Attention mechanism, which is designed to focus more on the cognitive process of learners. This feature makes the model pay more attention to the way learners think and the distribution of their attention while answering questions. In this case, the models rely more on contextual information and temporal attributes. As for the CSGKT model (including CSGKT_RKC and CSGKT_RC), it is a model that integrates multiple attributes. So the CSGKT model can capture learners' knowledge states more comprehensively in different contexts. By integrating information from different attributes, the model can better understand learners' learning processes, including their cognitive activities and knowledge states. This design of integrated attributes reflects the multifaceted considerations of the CSGKT model to improve the model's performance and adaptability, especially in learning environments with complex contexts and multiple influences. Furthermore, CSGKT_RC, which incorporates both context and time attributes, showed a 5.2% improvement in capturing the correlation between learners' adaptability and their knowledge state fluctuations compared to CSGKT_RC_RT. This integrated attribute design allows CSGKT to more accurately model how intrinsic characteristics influence knowledge states.

5.3. Model analysis

To further investigate the effectiveness of the four key components (schema formation using HGNN, association information formation using DGCNN, temporal modeling using LSTM, and cognitive stabilization using Erase-add Gate) of the CSGKT model, four variants of the CSGKT model were established. Their performance across different datasets is reported in Table 6.

- **CSGKT-RH** (Remove the forming schema component) For the CSGKT-RH variant, we do not utilize HGNN to extract schema for the extrinsic factors and intrinsic characteristics. The purpose of this variant is to evaluate the impact of using complex schema extracted by HGNN on the performance of CSGKT.
- **CSGKT-RD** (Remove the forming associated information component) For CSGKT-RD, we do not use DGCNN to extract the related information of concepts contained in the questions. The purpose of this variant is to evaluate the impact of using DGCNN to extract the dependency relationships between the concepts contained in the questions on the performance of CSGKT.
- **CSGKT-RL** (Remove the temporal modeling component): For CSGKT-RL, LSTM is replaced with a simple feed-forward neural network for temporal modeling. This variant aims to evaluate the role of LSTM in capturing long-term dependencies in the learning process.
- **CSGKT-RE** (Remove the stabilizing cognition component) For CSGKT-RE, we do not use the Erase-add Gate to filter out knowledge states that do not match the learner's current cognitive activity. The purpose of this variant is to evaluate the impact of using the Erase-add Gate to mitigate the bias introduced by self-training on the performance of CSGKT.

The various components of the CSGKT model play unique and important roles in the KT task. The experimental results in Table 6 demonstrate the contribution of each component to the model's performance.

The performance decline of CSGKT-RH is relatively small, especially on the Assist2017 and Statics2011 datasets, indicating that HGNN contributes to capturing learning environments and individual cognition. Although HGNN's impact is less pronounced than other components, its ability to process situational factors, coupled with its moderate computational complexity $O(m \cdot |E^z| \cdot d + n \cdot |E^f| \cdot d)$, remains an important part of the model.

CSGKT-RD shows the most significant performance drop, particularly on the Assist2017 dataset. This suggests that DGCNN plays a crucial role in modeling knowledge concept relationships, with its $O(k^2 \cdot d^2)$ complexity yielding substantial performance returns in practice.

CSGKT-RL exhibits a slight performance decrease, indicating that LSTM's temporal modeling capability contributes to capturing long-term learning patterns. Although LSTM's contribution is relatively small, its scalability when handling long sequences (complexity $O(|S| \cdot d^2)$) makes it an indispensable part of the model.

The performance reduction of CSGKT-RE falls between that of CSGKT-RH and CSGKT-RD, indicating that the Erase-add Gate plays an important role in aligning cognitive activities and knowledge states. Given its low computational complexity $O(d)$, the Erase-add Gate demonstrates an excellent balance between performance and complexity.

Overall, the total time complexity of the CSGKT model can be approximated as $O(m \cdot |E^z| \cdot d + n \cdot |E^f| \cdot d + k^2 \cdot d^2 + |S| \cdot d^2 + d)$. This reflects the computational challenges the model faces when processing large-scale data. Computational efficiency is affected, particularly when situational factors are complex (large m , n), knowledge concepts are numerous (large k), or learning sequences are long (large $|S|$).

The performance improvement of the CSGKT model is not only reflected in data metrics but, more importantly, provides technical support for validating two key cognitive psychology assertions. Firstly, DGCNN lays the foundation for verifying the assertion that "short-term memory forgetting of knowledge concepts is mainly caused by interference rather than memory trace fading". The significant performance decline of CSGKT-RD indicates that when the model cannot effectively handle interference between concepts, prediction accuracy decreases substantially. This emphasizes the importance of considering inter-concept interactions in KT, supporting the crucial role of interference theory in short-term memory forgetting.

Secondly, HGNN provides technical support for validating the assertion that "cognitive imagery and perceptual function play an equivalent role in the cognitive process". The model innovatively utilizes HGNN to simultaneously capture intrinsic characteristics and extrinsic factors. This design can be used to simulate the different influences of cognitive representations (intrinsic, variable mental representations) and perceptual functions (extrinsic, relatively stable perceptual processing) in the learning process. The relatively small performance change of CSGKT-RH supports, from a contribution perspective, the view that cognitive representations and perceptual functions have similar importance in cognitive processes.

In the following sections, the impact of the forming schema component, the forming associated information component, and the stabilizing cognition component on the model are analyzed through visualization.

5.3.1. Effect of forming schema

Fig. 6 illustrates the association networks of extrinsic factors and intrinsic characteristics in the Assist2009 and KDDCup2010 datasets. It reveals the multidimensional nature of cognitive situations, learning environments, and the interactions of learners' characteristics.

Regarding extrinsic factors, strong connections are observed among "Question", "Assignment", and "Teacher", reflecting the core components of educational activities. Notably, "Tutor" is closely linked

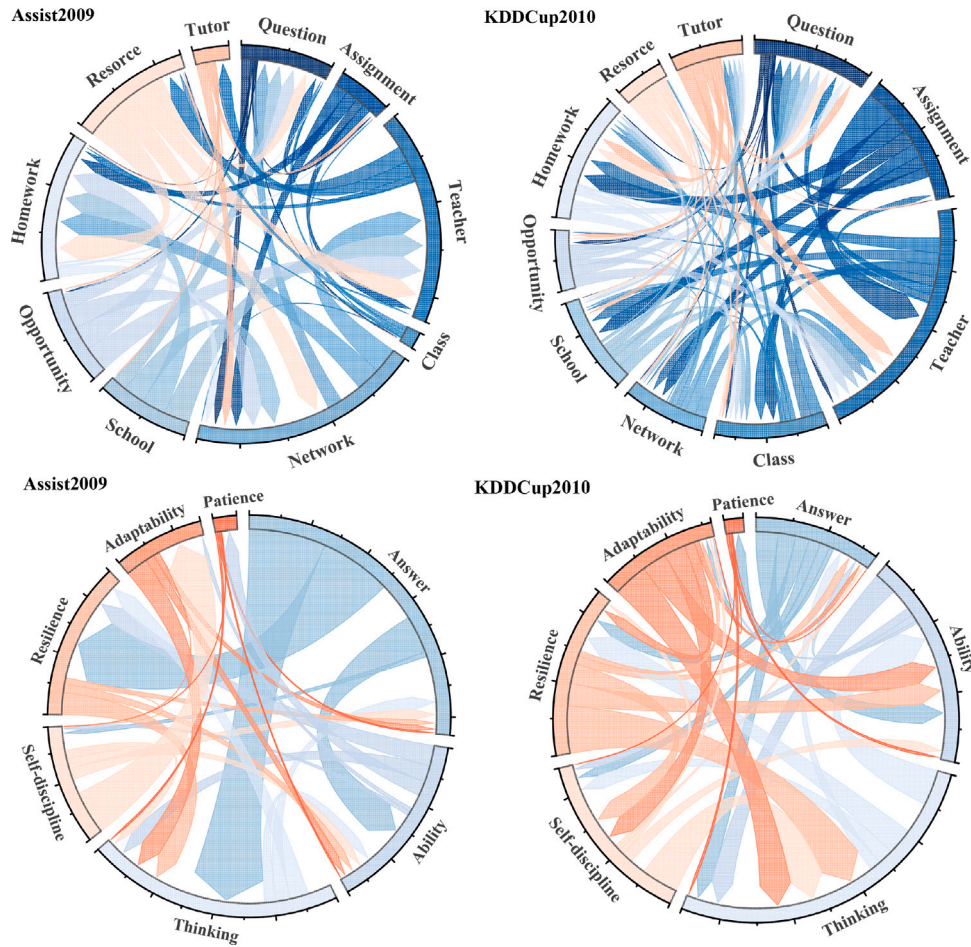


Fig. 6. Partial schema of extrinsic factors and intrinsic characteristics.

to multiple other factors, indicating the crucial role of personalized tutoring in the learning process. The connection between “Resource” and “Opportunity” suggests that the availability of learning resources influences the allocation of learning opportunities.

The visualization of intrinsic characteristics reveals complex interactions among learners’ traits. Strong connections between “Answer”, “Ability”, and “Thinking” indicate that these three factors are key indicators for predicting learning performance. The link between “Self-discipline” and “Patience” reflects how learners’ personality traits influence their learning behavior. It is noteworthy that “Adaptability” shows extensive connections with multiple factors in both datasets, implying that learners’ adaptability plays a significant role across different learning environments.

The figure demonstrates that the learning environment’s influence on cognitive processes involves numerous factors with intricately interconnected schemas, while learners’ intrinsic characteristics, though fewer, also exhibit interrelations. The extrinsic influences on learners vary in different environments, i.e., across different datasets. For instance, the “Network” factor in the KDDCup2010 dataset shows denser connections compared to Assist2009, reflecting the importance of online learning environments in that dataset. These observations align with the concept of cognitive situations.

This in-depth schema analysis, utilizing HGNN, provides a crucial foundation for improving prediction accuracy and understanding the influence of intrinsic characteristics on knowledge states. The schema structure’s ability to capture these influences is particularly noteworthy. For instance, the strong connection between “Thinking” and “Answer” reveals how variations in thinking patterns lead to different knowledge state trajectories. This nuanced understanding allows for

more accurate predictions of how individual differences in intrinsic characteristics shape learning outcomes over time. By capturing the interplay between extrinsic factors and intrinsic characteristics, the model offers a comprehensive representation of cognitive states, dynamically adjusting factor weights based on observed association strengths in different datasets. This approach not only improves the accuracy of learning behavior predictions but also provides valuable guidance for designing effective, personalized teaching strategies that account for the complex interaction between learners’ intrinsic characteristics and their evolving knowledge states.

5.3.2. Effect of forming associated information

To visually analyze the directionality and correlation of nodes, we randomly selected ten knowledge concepts from the Assist2009 dataset (‘Rounding’, ‘Multiplication and Division Integers’, ‘Ordering Integers’, ‘Surface Area Rectangular Prism’, ‘Addition and Subtraction Integers’, ‘Perimeter of a Polygon’, ‘Equation Solving more than Two Steps’, ‘Least Common Multiple’, ‘Multiplication Fractions’, and ‘Pythagorean Theorem’). Using DGCNN, each concept is decoupled into in-degree and out-degree and projected onto a two-dimensional space, as shown in Fig. 7. The x and y coordinates represent the proportion of in-degree and out-degree to the total degree of the node, respectively.

Fig. 7 reveals a dispersed and irregular distribution pattern of knowledge concepts, reflecting the complexity of knowledge structures. Specifically, concepts such as ‘Addition and Subtraction Integers’, ‘Rounding’, and ‘Ordering Integers’ exhibit larger out-degree proportions. These concepts are almost distributed along the $(0, y)$ line, indicating their role as foundational knowledge with strong forward-guiding characteristics. Conversely, concepts like ‘Equation Solving

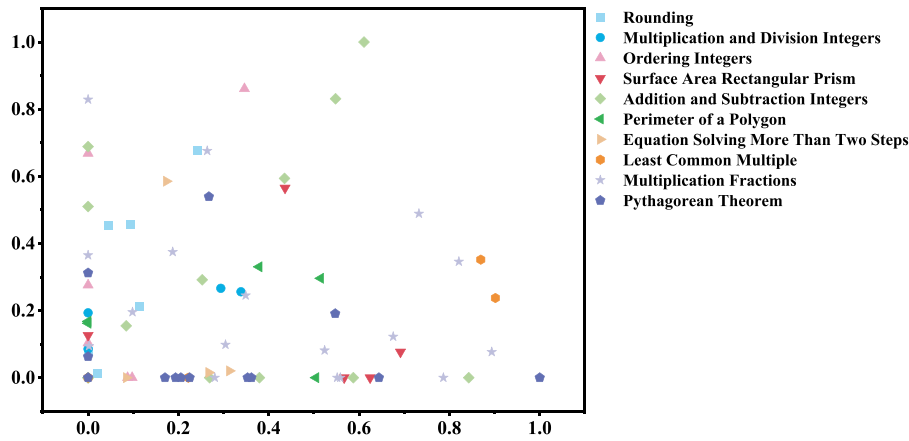


Fig. 7. The association structure between some knowledge concept nodes.

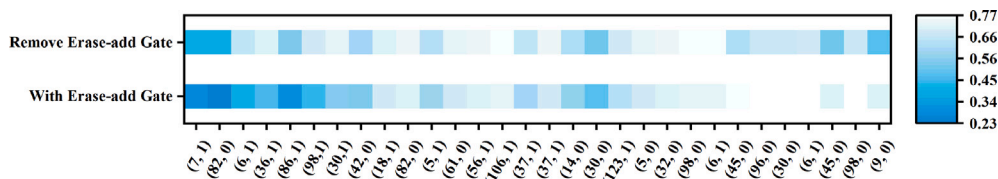


Fig. 8. The impact of the Erase-add Gate on prediction results.

More Than Two Steps', 'Surface Area Rectangular Prism', and 'Pythagorean Theorem' show higher in-degree proportions, clustered near the $(x, 0)$ line. This aligns with their nature as theorems and problem-solving steps in geometry and algebra, often dependent on fundamental knowledge. Concepts scattered within the quadrants indicate their interdependence with other concepts.

Notably, the group of mathematical concepts including 'Multiplication and Division Integers', 'Ordering Integers', and 'Addition and Subtraction Integers' exhibits similar in-degree distributions but significant differences in out-degree. This suggests similar prerequisite knowledge but varying instructional roles, consistent with the basic usage patterns of integer concepts. By observing the in-degree and out-degree distributions of different concepts, reasonable inferences can be made about their dependencies and logical order within the knowledge system.

DGCNN demonstrates excellent performance in processing concept relationship networks, precisely capturing hierarchical relationships and dependencies between concepts. By identifying the relative importance of concepts and their nuanced relationships, DGCNN enhances the model's understanding and predictive capability of learners' cognitive states. This approach not only improves overall prediction accuracy but also aids in identifying potential learning bottlenecks and key concepts. Consequently, educators can design personalized learning paths based on this structured information and develop more targeted teaching strategies at critical learning points.

5.3.3. Effect of stabilizing cognition

We sequentially selected thirty questions and their answers from the Assist2009 dataset. Fig. 8 shows the prediction results of the CSGKT-RE model and the CSGKT model after 50 iterations. This heatmap presents the prediction patterns of the models with and without the Erase-add Gate on a continuous sequence of questions.

It can be visually observed that as questions change, such as from $(6, 1)$ to $(9, 0)$, although the color gradients of corresponding blocks differ between the two models, the degree of change in color gradients between adjacent blocks is similar, indicating consistency in the fluctuation of prediction results. However, the CSGKT model exhibits noticeably more stable fluctuations, as seen in the range $(7, 1)$ to

$(42, 0)$. The CSGKT-RE model shows more abrupt jumps in changes, particularly in the ranges $(82, 0)$ to $(6, 1)$ and $(30, 1)$ to $(42, 0)$.

This comparison visually demonstrates the effectiveness of the Erase-add Gate in maintaining prediction stability. The stabilizing effect of the Erase-add Gate contributes to improved prediction accuracy in several key aspects: Firstly, it smooths out short-term anomalous fluctuations, preventing the model from overreacting to temporary changes in cognitive states. Secondly, it enables the model to better capture the gradual changes in learners' cognitive states, which aligns more closely with the characteristics of knowledge accumulation in actual learning processes. Finally, this stability mechanism likely helps the model maintain robustness when faced with noisy data, such as when learners occasionally provide results inconsistent with their current knowledge state. This not only enhances the overall prediction accuracy of the model but also provides educators with a more stable assessment of learners' cognitive states, facilitating the development of more coherent and effective teaching strategies.

6. Cognitive psychology validation experiment

The KT task can reflect subtle changes in learners' cognitive processes, and the results help explore the degree of deviation that different individuals produce in the mean process. We conduct the following two cognitive psychology validation experiments.

6.1. Short-term memory forgetting of knowledge concepts is mainly caused by interference rather than memory trace fading

During the cognitive process, learners engage in information processing, comparison, and judgment, and must concentrate on this task. This process involves various sources of interference information [61], including task interference (other tasks or information interfering with the current task) and intrinsic interference (such as emotions and pressure). These interferences can cause deviations in learners' cognitive processes, leading to reduced acquisition of knowledge concepts and subsequent impacts on their learning performance. On the other hand, memory trace fading [62] can also affect long-term learning

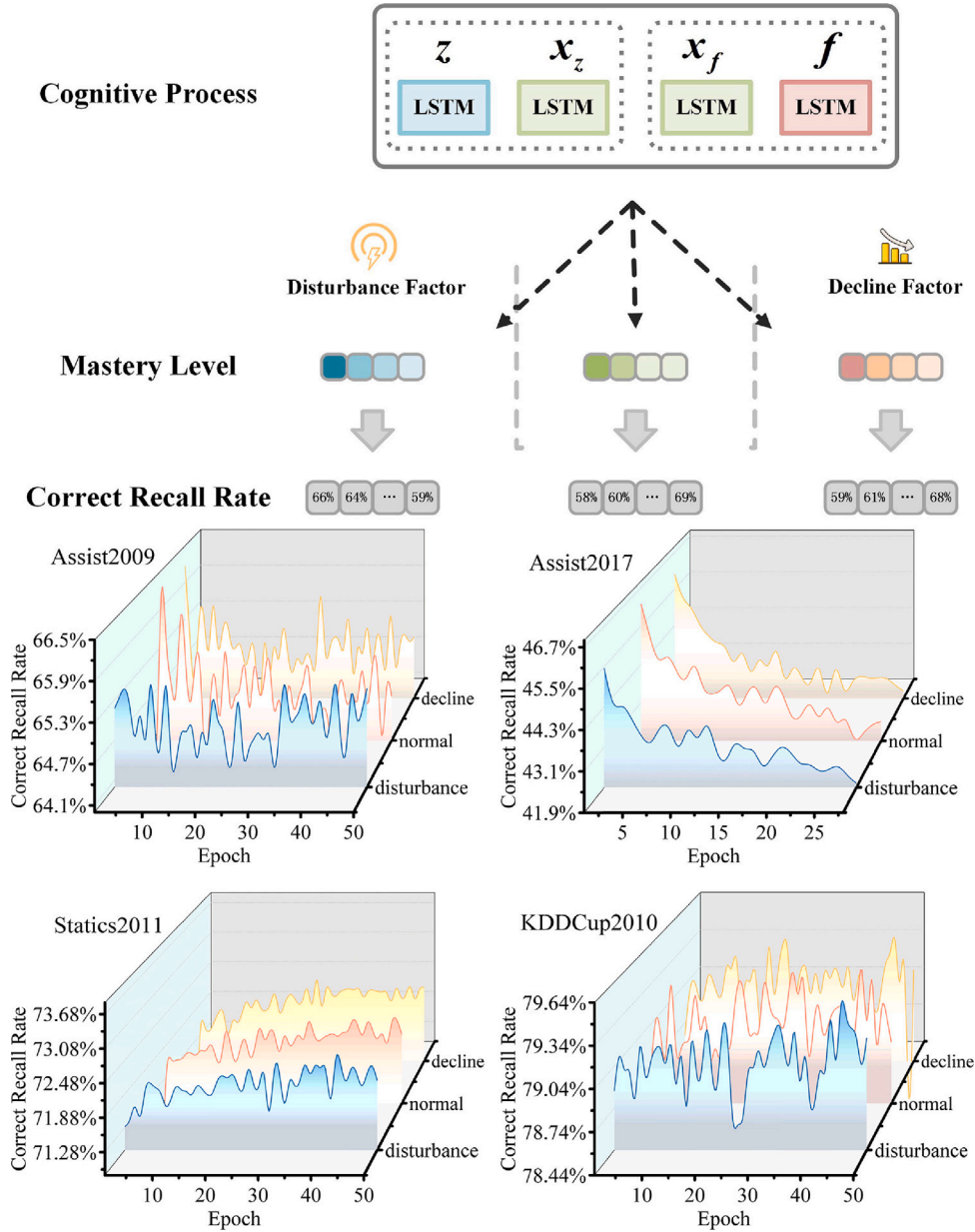


Fig. 9. Introducing interference factor ρ_{int} or fading factor ρ_{fad} in the cognitive process, and training to obtain the learner's phased performance, the waterfall chart at the bottom reflects the situation of being affected by interference and memory trace fading, and the CRR of knowledge concepts at each stage.

performance, as learners have difficulty recalling previously learned knowledge concepts over time.

In our model, the DGCNN component is designed to simulate the interactions between knowledge concepts, including potential interference effects. Specifically, the dynamic connection weights between concept nodes in DGCNN reflect the intensity of interactions among different concepts. This design enables the model to capture potential interference relationships between concepts.

Fig. 9 illustrates the impact of interference factor ρ_{int} or fading factor ρ_{fad} on learners' knowledge acquisition concepts during the cognitive process, which is expressed in Eqs. (15) and (16) respectively.

$$\begin{aligned} x_{t+1}^z &= \text{LSTM}(x_t^z \rho_{\text{int}}) \\ x_{t+1}^f &= \text{LSTM}(x_t^f \rho_{\text{int}}) \end{aligned} \quad (15)$$

$$\begin{aligned} x_{t+1}^z &= \text{LSTM}(x_t^z \rho_{\text{fad}}) \\ x_{t+1}^f &= \text{LSTM}(x_t^f \rho_{\text{fad}}) \end{aligned} \quad (16)$$

where $\text{LSTM}(\cdot)$ is calculated in the same way as Eq. (9). At each time stamp, the interference factor or fading factor affects the hidden state. The correct recall rate (CRR) of learners was measured. The CRR varies with the number of correctly recalled knowledge concepts (NCR) and the total number of knowledge concepts (TN). The CRR under different conditions can be calculated as shown in Eq. (17).

$$\text{CRR} = (\text{NCR}/\text{TN}) \times 100\% \quad (17)$$

The waterfall chart at the bottom of Fig. 9 clearly shows that the degree of fluctuation in CRR under the influence of memory trace fading is highly consistent with the normal situation, while the degree of fluctuation in CRR under the influence of interference is stronger than that under the influence of memory trace fading.

This result aligns with our model design. The DGCNN, through its dynamic graph structure, updates the relationship weights between concepts at each time step, enabling the model to respond more sensitively to fluctuations caused by interference factors. In contrast, the LSTM structure exhibits a smoother trend when simulating memory

Table 7

Deviation values of cognitive imagery and perceptual function on different datasets.

Datasets	Assist2009		Assist2017		Statics2011		KDDCup2010	
	Perceptual Function	Cognitive Imagery	Perceptual Function	Cognitive Imagery	Perceptual Function	Cognitive Imagery	Perceptual Function	Cognitive Imagery
1	0.0056	<u>0.0113</u>	0.0006	0.0007	0.0009	0.0022	0.0013	0.0021
2	0	0.0067	0.0021	0.0025	0.0015	0.0013	0.0005	0.0005
3	0.0064	0.0094	0.0007	0.0016	0.0008	0.0002	0.0002	0.0007
4	0.0004	0.001	0.0006	0.0003	0.0027	0.0021	0.0001	0.0015
5	<u>0.0088</u>	0.0077	0.0049	0.0005	0.0031	0.0032	0.0009	0.0004
6	0.0017	0.0056	0.0042	0.0004	0.0028	<u>0.0038</u>	0.0002	0.0005
7	0.003	0.0079	0.0041	0.0006	0.0029	0.0018	0.0002	0.0011
8	0.0036	0.0053	0.0023	0.0045	<u>0.0033</u>	0.0025	0.0012	0.0015
9	0.0027	0.007	<u>0.0134</u>	0.0012	0.0013	0.0023	0.0002	0.0028
10	0.005	0.0028	0.0009	0.0049	0.0009	0.0007	0.0002	0.0007
11	0.008	0.0107	0.0006	<u>0.0059</u>	1E-04	0.0016	0.0021	0.0007
12	0.0031	0.0025	0.0013	0.0053	0.0027	0.001	0.0002	0.0016
13	0.002	0.0051	0.0048	0.0039	0.002	0.001	0.0008	0.0021
14	0.0041	0.002	0.0021	0.0058	0.0023	0.0002	0.0011	0.0005
15	0.0055	0.0017	0.002	0.0018	0.0007	1E-04	<u>0.0028</u>	<u>0.0043</u>

decay. This difference reflects the model's ability to distinguish between interference effects and natural memory decay, consistent with the observed experimental phenomena.

The experimental results not only support the theory that short-term memory forgetting is primarily caused by interference but also validate the effectiveness of our model in simulating different types of cognitive processes. This finding emphasizes the importance of considering inter-concept interference in KT tasks, providing direction for further optimization of model structures to more accurately predict learning states.

6.2. Cognitive imagery and perceptual function play an equivalent role in the cognitive process

Cognitive imagery [63,64] refers to the mental intrinsic representation generated by learners through intrinsic thinking and imagination, which is flexible and variable in terms of manipulation and change. On the other hand, the perceptual function [65,66] denotes the learners' ability to perceive the extrinsic world and convert extrinsic perceptual information into neural signals. As perceptual function passively receives and processes extrinsic information, it remains stable and invariant. The study simulates the effects of cognitive imagery and perceptual function on the cognitive process by controlling intrinsic characteristics and extrinsic factors.

In our model, the HGNN component is designed to simultaneously process intrinsic characteristics and extrinsic factors. Intrinsic characteristics can be analogized to cognitive imagery, reflecting learners' intrinsic cognitive levels, while extrinsic factors are analogous to perceptual functions, representing extrinsic stimuli in the learning environment. By integrating these two types of information within the same graph structure, HGNN provides a technical foundation for validating the equivalent roles of cognitive imagery and perceptual functions.

To more precisely simulate the roles of cognitive imagery and perceptual function, we have implemented a fine-grained masking method. Specifically, we introduce masking matrices M_f and M_z , which have dimensions identical to the intrinsic characteristics vector f_i^l and extrinsic factor vector z_i^l , respectively. These masking matrices contain elements of 0 or 1, enabling selective masking of specific characteristics or factors through element-wise multiplication ($\odot$). For masking cognitive imagery (intrinsic characteristics), we employ the following formula:

$$f_i^{l+1} = \delta((D_f^v)^{-1/2} H_f W_f (D_f^e)^{-1/2} (H_f)^T (D_f^v)^{-1/2} (f_i^l \odot M_f) \Omega_i) \quad (18)$$

Similarly, for masking perceptual function (extrinsic factors), we utilize:

$$z_i^{l+1} = \delta((D_z^v)^{-1/2} H_z W_z (D_z^e)^{-1/2} (H_z)^T (D_z^v)^{-1/2} (z_i^l \odot M_z) \Theta_i) \quad (19)$$

This methodology allows for flexible control over the masking process. For instance, given an n -dimensional intrinsic characteristics vector f_i^l , we can set $M_f = [0, 1, 1, \dots, 1]^T$ to mask the first characteristic while preserving others. Similarly, we can configure various patterns for M_z to mask extrinsic factors. This fine-grained masking approach enables us to more accurately model scenarios where cognitive imagery and perceptual function are partially absent or diminished in real-world settings, thereby better reflecting the complexities of authentic learning environments.

The experimental results strongly support the theory of equivalent roles of cognitive imagery and perceptual functions in cognitive processes. The kite diagram in Fig. 10 demonstrates that concealing intrinsic characteristics (simulating cognitive imagery) and extrinsic factors (simulating perceptual functions) exhibit highly similar fluctuation patterns in their influence on learners' mastery of knowledge concepts. This similarity indicates that both intrinsic cognitive imagery and extrinsic perceptual inputs impact the learning process. Table 7 further quantifies this similarity, showing that the deviation values between cognitive imagery and perceptual functions remain at low levels across all tested datasets, with the maximum deviation not exceeding 0.015. This small deviation statistically supports the equivalence of their roles. Additionally, HGNN allows dynamic interactions between intrinsic characteristics and extrinsic factors within the graph structure, simulating the complex interplay between intrinsic cognition and extrinsic stimuli in the learning process. Experimental results show that even when one type of information is concealed, the model maintains relatively stable performance, demonstrating its balanced handling of cognitive imagery and perceptual functions. Notably, this result shows consistency across multiple datasets (Assist2009, Assist2017, Statics2011, KDDCup2010), enhancing the reliability and universality of the conclusion.

These findings not only validate important theories in cognitive psychology but also provide crucial insights for the design of KT models. They emphasize the necessity of equally considering learners' intrinsic cognitive levels and extrinsic learning environments in model design, pointing the way for creating more comprehensive and effective KT systems. Through this approach, our model not only achieves precise simulation of cognitive processes technically but also provides empirical support for the equivalence of cognitive imagery and perceptual functions at a theoretical level.

7. Conclusions

In this study, a cognitive situation-based graph knowledge tracing model is proposed to solve the problem that traditional methods ignore the influence of learners' intrinsic cognitive characteristics and extrinsic environmental factors on cognitive activities and knowledge state

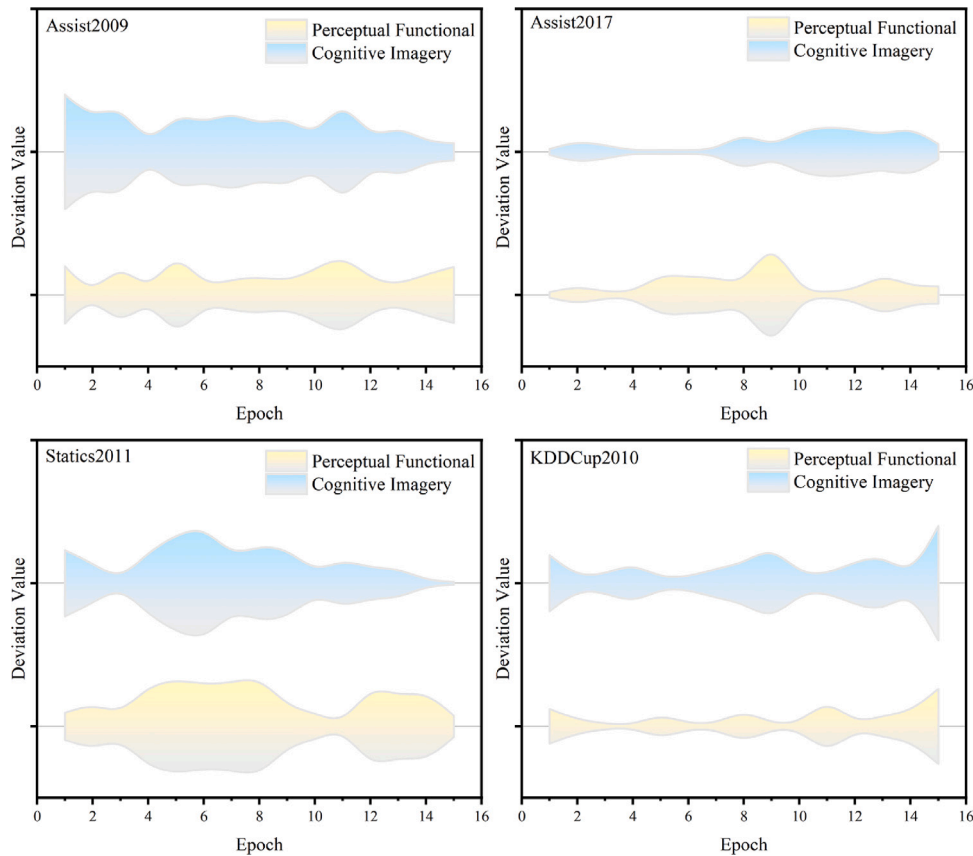


Fig. 10. The role of cognitive imagery and perceptual function in the cognitive process was validated through a kite plot.

changes. CSGKT constructs a correlation analysis model of extrinsic factors, cognitive processes, and intrinsic characteristics with the help of cognitive psychology theories, and improves the ability to describe complex learning environments. CSGKT uses HGNN and DGCNN to model the influence and knowledge structure of learners, LSTM for state tracing, and Erase-add Gate for cognitive adjustment and realizes the simulation of dual state changes under the dual influence of learners.

Through training and testing on four real datasets, the results show that the model's KT performance is significantly better than that of the traditional deep learning model, the attention-based model, and the graph-based model, and realizes the prediction of state objects. The analysis of the effectiveness of the model on each component shows that the modeling of extrinsic factors and intrinsic characteristics, the learning of knowledge associations, and the cognitive regulation mechanism all have a significant effect on the performance of the model. The simulation of two major experiments in cognitive psychology verifies that the main cause of short-term memory forgetting of knowledge concepts is interference rather than memory trace fading and that cognitive imagery and perceptual function are equivalent to cognitive processes, indicating that the model can reflect cognitive laws.

In terms of time complexity, this method extracts features and association information through parallel HGNN and DGCNN and then trains them through LSTM, whose time complexity is $O(n)$, which is better than the traditional RNN method with serial training. In terms of spatial complexity, this method constructs an overall associated dual-state graph structure to express the learning process, which is more comprehensive than the single-state expression, but the consumption of computational resources is correspondingly larger. From a comprehensive point of view, the proposed method can control time consumption through parallel training while guaranteeing the expression ability, which has certain advantages of time and space efficiency.

CSGKT demonstrates excellent performance across datasets from various disciplines such as mathematics and engineering, indicating

its potential for broad application in diverse educational domains. The model's core components (HGNN, DGCNN, LSTM, and Erase-add Gate) exhibit strong generalizability, capable of adapting to different types of learning data and knowledge structures. However, some adjustments are necessary when applying the model to new fields. These adjustments include modifying feature definitions based on cognitive situations, optimizing knowledge structure representations, adjusting time scales, considering computational resource limitations, and adapting to varying data quality and quantity. Despite these potential adjustment needs, the fundamental framework and principles of CSGKT should apply to a wide range of educational domains.

Moreover, CSGKT is not merely a prediction tool but can also provide rich information for adaptive intervention strategies. By analyzing the outputs of various model components, personalized feedback and instructional adjustment strategies can be designed. For instance, the output from HGNN can reveal interactions between different learning factors, aiding in the construction of comprehensive teaching intervention strategies, including adjustments to teaching pace, matching of learning partners, and selection of teaching methods. The output from DGCNN can be used to analyze learners' transfer abilities between different knowledge concepts, providing guidance for designing knowledge integration and application exercises. The output from the Erase-Add Gate reflects the gap between learners' current cognitive activities and ideal knowledge states, offering a scientific basis for formulating personalized learning methods.

8. Discussion

8.1. Theoretical contributions and practical implications

This study significantly advances the field of KT across theoretical, methodological, and practical dimensions through the introduction of the cognitive situation concept and the CSGKT model.

Theoretically, the cognitive situation concept expands the KT theoretical framework. This innovative concept integrates learners' extrinsic factors, intrinsic characteristics, and cognitive processes, offering a multidimensional perspective for KT tasks. The CSGKT model, by simultaneously considering environmental influences and individual cognitive characteristics, addresses the limitations of traditional KT methods that often neglect the complexity of learning environments. This approach facilitates a shift in KT research from static knowledge assessment to dynamic learning process analysis.

Methodologically, the CSGKT model innovatively combines advanced algorithms such as HGNN, DGCNN, LSTM, and Erase-add Gate, achieving effective modeling of complex learning environments. This integrated approach not only enhances predictive performance but also improves model interpretability. Notably, the simulation of two cognitive psychology assertions provides new directions for future KT research, potentially facilitating more accurate learning state assessments and the development of personalized instructional strategies.

In terms of application, the CSGKT model offers crucial guidance for designing intelligent educational systems. Its multidimensional analytical capability enables dynamic adjustments of learning content difficulty and presentation modes, adapting to learners' cognitive states and environmental factors. This real-time adaptive capacity significantly enhances the practical value of KT in educational settings, promoting a transition towards truly personalized learning experiences in intelligent education.

Moreover, this research strengthens interdisciplinary connections between KT and cognitive science by validating classic cognitive psychology hypotheses. This theoretical validation not only reinforces the scientific foundation of the CSGKT model but also indicates future research directions for integrating cognitive theories with machine learning techniques. This interdisciplinary approach has the potential to propel KT beyond mere performance prediction towards a deeper understanding of learning mechanisms, providing more robust theoretical support for educational practices.

8.2. Limitations and future research directions

Despite the significant advancements made in cognitive modeling and KT, this study has several limitations that require improvement and expansion in future research.

Firstly, while this study focuses on extrinsic factors, intrinsic characteristics, and cognitive processes within cognitive situations, it lacks sufficient exploration of cognitive load issues in the learning process. According to cognitive load theory, learners have limited working memory capacity when processing information. When the complexity of learning tasks exceeds the learner's processing ability, excessive cognitive load can negatively impact learning and memory. Future research endeavors could further refine cognitive load theory and develop dynamic cognitive load estimation algorithms within the CSGKT framework. These advancements could leverage physiological measurement data to provide real-time cognitive load assessments. Such innovations may potentially enhance intelligent educational systems, enabling automatic adjustments of task difficulty and presentation modalities based on learners' current cognitive load. This adaptive approach would aim to ensure optimal challenge levels and maximize learning efficiency.

Secondly, this study primarily concentrates on multi-perspective KT within a single domain, with insufficient attention to cross-domain or cross-task knowledge transfer issues. Learning transfer, the ability of learners to apply acquired knowledge and skills to new situations or questions, is key to effective learning. Future research could explore how to utilize the CSGKT model to support knowledge transfer, such as introducing transfer learning techniques to enhance model adaptability to new learners or domains through cross-domain knowledge mapping and sharing. Additionally, the role of metacognitive strategies in knowledge transfer could be investigated to help learners better monitor and regulate their learning processes, promoting deeper understanding

and application. We propose evaluating the CSGKT model's efficacy in predicting knowledge transfer across related yet distinct domains, such as the transition from algebraic to geometric mathematical concepts. This investigation may entail developing a multi-task learning architecture within the CSGKT framework, simultaneously modeling and predicting performance across multiple related domains. Such an approach has the potential to explain shared cognitive processes and knowledge structures that facilitate transfer.

Furthermore, this study mainly models learner behavior data, with limited consideration of emotional factors. Learners' emotional states are closely related to cognitive engagement, with positive emotions facilitating learning motivation and attention, while negative emotions may lead to dispersed cognitive resources and decreased learning effectiveness. Future research could consider integrating emotional data, such as facial expressions and voice intonation, to enhance a comprehensive understanding of learners' cognitive-emotional states, providing a basis for emotional regulation and personalized interventions. Future research endeavors could integrate multimodal affect recognition techniques into the CSGKT model, incorporating facial expression analysis, vocal emotion analysis, and even physiological signals such as heart rate variability. By implementing emotion-sensitive pedagogical agents or adaptive affective support strategies, we can enhance the efficacy of intelligent educational systems. This approach aims to provide learners with more personalized and emotionally adaptive learning experiences, ultimately improving learning outcomes and engagement.

Lastly, this study, based on cognitive psychology theories, preliminarily explores the possibility of combining them with computational models. Future research could further explore potential avenues for these two cognitive psychology principles. Regarding the theory that short-term memory forgetting is primarily caused by interference, more refined experiments could be designed to distinguish the effects of different types of interference (e.g., task-related and emotional interference) on memory. This may involve introducing more complex interference mechanisms into the model. For the study of equivalent roles of cognitive imagery and perceptual functions, we could explore whether this equivalence holds across different learning tasks and environments. For instance, experiments could be designed to compare the relative importance of intrinsic cognition and extrinsic stimuli in abstract concept learning versus concrete skill acquisition processes. Moreover, by introducing more diverse learner characteristics and environmental factors, we could construct a more comprehensive HGNN approach to capture the dynamic balance between cognitive imagery and perceptual functions in complex learning scenarios. These studies could not only deepen our understanding of cognitive processes but also potentially provide more precise theoretical guidance for the design of personalized learning systems.

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CRediT authorship contribution statement

Qianxi Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Weidong Ji:** Funding acquisition, Conceptualization. **Guohui Zhou:** Conceptualization. **Yingchun Yang:** Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ji Weidong reports was provided by Harbin Normal University.

Data availability

Data will be made available on request.

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